
SCI-RTC

Paired Raw-Timestream Conditioning and Temporal Response

Engineering Conformance View

Package version	v0.1
Revision	r0.12
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Authority	Shared normative core in <code>src/common/</code>
View-specific content	Informative evidence and review guidance only
Status	Scientific authority frozen by owner; implementation conformity not assessed under this contract.

Claim boundary. This view carries the frozen scientific authority in the shared core. It does not assert implementation conformity, representation fidelity, validation, observational performance, or production readiness. It contains no independent normative science.
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How to use this view

The six-file shared normative core below is imported exactly once and is the complete formal authority companion to the separate twelve-section science-team rationale. The rationale explains this authority but does not duplicate it. The material after the end-of-core marker is an informative method for recording evidence against that authority. It may not waive a requirement, select an owner decision, redefine an unavailable state, or restate an equation with different semantics.

A conformance review should create one evidence record per requirement and one falsification record per applicable prediction. Each record must bind an exact implementation and artifact identity, state what was inspected or executed, preserve unexpected error-level messages, and distinguish pass, fail, not exercised, blocked by owner decision, and unavailable evidence. Passing a fixture supports only the named requirement and claim layer.

Stop rule. If a required owner decision is open, the affected operation or claim remains unavailable. Engineering must not choose a numerical value from current behavior and then describe it as scientific authority.

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1 Shared normative authority and notation

Normative core stamp: SCI-RTC-v0.1/r0.12. The six files in `src/common/` form one authority. Text outside this shared core is explanatory or conformance guidance and cannot alter a definition, equation, assumption, requirement, prediction, unavailable state, or owner decision.

SCI-RTC begins with admitted aligned paired raw detector-coordinate streams **xs** and **rs** and ends with an atomic, consumer-neutral raw conditioning bundle. Conditioned x is the required numerical RTC output; when conditioned r is requested, it is the optional paired companion on exactly the conditioned- x grid. The exact paired raw r parent and every causal r diagnostic or selector remain in that bundle whether or not the companion is requested. Conditioned r is neither an independent RTC pipeline nor a calibrated or Stokes product. Raw x is not itself a Stokes parameter. ALIGN, the Tune/readout mapping authority, AST, and SCI-BEAM retain authority for the upstream meanings they produce; SCI-CAL may calibrate only the conditioned x path of an exact RTC bundle. PTC, VAL, MAP, BEAM, NOI, MAP-003, FLT, SRC/MODE, and FRUIT retain authority for their downstream science. This contract owns the realized conditioning order and its scientific consequences; it does not infer current software behavior.

The words **shall** and **shall not** identify requirements. **Required unavailable** means that the affected product or claim shall not be reported as complete. **Optional unavailable** is a typed, scientifically inert state. Missing, disabled, automatic, rejected, invalid, failed, unavailable, unsupported, and omitted remain distinct states.

1.1 Indices, sets, and application-context labels

Symbol	Normative meaning
o	Immutable observation identity.
$d, q \in \mathcal{D}_o$	Exact admitted detector occurrences; d is a target and q may be a donor. A detector index or row number alone is not this identity.
m_{native}	Native acquisition-row index; it is not a physical r coordinate or an aligned or conditioned slot.
$j \in \{0, \dots, N-1\}$	Ordered aligned input slot on the admitted ALIGN grid.
$n \in \{0, \dots, N_y-1\}$	Ordered phase-zero RTC output slot.
$\mathcal{R}$	RTC application context naming requested use, paired input and interval, consumers, and constraints. Pointing, OOF, Science, Beammap, and diagnostic labels may be context facts; no such label restricts operation availability or silently selects policy.
$x_{dm}^{\text{acq}}, r_{dm}^{\text{acq}}$	Exactly paired raw readout coordinates on the native acquisition grid, before the imported ALIGN mapping.
x_{dj}^A, r_{dj}^A	Exactly paired admitted aligned raw KID readout coordinates and the RTC-local input boundary. x supplies the ordinary total-intensity analysis path; r supplies readout/leakage/artifact/Tune-health evidence. They share occurrence identity, cadence, phase, and mapping lineage but retain independent value validity.
$\mathbf{v}_{dm}^{IQ, \text{acq}}$	Upstream native-grid orthogonal microwave transmission components $(I_{dm}^{\text{acq}}, Q_{dm}^{\text{acq}})^T$.
$\mathbf{u}_{dj}^{xr}$	Paired raw readout vector $(x_{dj}^A, r_{dj}^A)^T$ admitted at the RTC boundary.
ζ	Complete Tune and $I, Q \rightarrow x, r$ mapping identity, reference, normalization, validity domain, epoch, and uncertainty state.
χ_{dj}	Detector-static flxscale comparison coordinate used only to define compatible donor transfer; it is not a calibrated RTC product.
u_{dj}^x	Raw x -coordinate $\Delta f/f$ value immediately before the ordered RTC filter chain.
$v_{dj}^{x, \text{preD}}$	Final raw x stream after every admitted RTC replacement/filter stage and immediately before phase-zero point selection.
$y_{dn}^{x, \text{RTC}}$	Required conditioned raw x -coordinate $\Delta f/f$ RTC output at the selected phase-zero slot.
$y_{dn}^{r, \text{RTC}}$	Optional conditioned raw r -coordinate companion at exactly the paired conditioned- x slot. It retains the raw- r coordinate convention and is typed invalid or unavailable where no scientifically justified r value exists.
c_{dn}^{CAL}	Distinct downstream SCI-CAL output, when authorized: top-of-atmosphere, point-source-equivalent mJy per fixed nominal beam.
ϕ_d	flxscale for one exact detector occurrence under the declared multiplicative convention and calibration domain.
$A_\alpha, \mathbf{a}_\alpha$	Imported, realized upstream ALIGN mapping and affine term with state α . They map the native acquired pair to the admitted aligned pair, remain in lineage, and are not reapplied by the RTC-local operator.
C_κ^{CAL}	Downstream SCI-CAL operator with state κ , including selected absolute flxscale and target-atmosphere correction; RTC does not apply it.
S_σ	Spike-candidate evidence selector used only for masking or robust downweighting during level-shift learning; it does not replace samples.
B_ω^x	Realized post-segmentation x -replacement operator conditioned on selection state ω ; it has no r numerical branch.
F_{k, θ_k}	Realized ordered FIR, IIR, notch, or admitted mask-controlled stage k .
$D_{M,0}$	Integer phase-zero point selector of factor $M \geq 1$.

Symbol	Normative meaning (continued)
Ω	Complete realized conditioning state, including all upstream bindings and every selected stage, exception, state, edge rule, and output interval.
$\mathcal{L}$	Declared learning population with exact identity, selection, support, masks, exclusions, and compatibility domain.
Λ	Learned evidence and uncertainty produced by declared diagnostics; it is not authorization to apply.
E_x, E_r, E_{xr}	Direct coordinate-specific and genuinely joint evidence records, preserving origin, support, protection, uncertainty, quality, causal class, and disposition.
B_{pair}	Cause-preserving union of accepted owner-designated hard action support from x , r , and joint evidence.
Π	One immutable resolved RTC pair plan admitted from learned evidence and fixed scientific policy.
L_Π	One ordinary temporal and sampling operator used identically on both canonical coordinate branches.
$\mathcal{A}_\Pi$	Apply operator that consumes plan Π without changing it.
$a \in \{1, \dots, A\}$	Refinement-attempt index; A is the number of attempts actually executed.
A_{max}	Finite owner-approved maximum attempt count, with $A \leq A_{\text{max}} < \infty$.
$k \in \{0, \dots, K\}$	Accepted-plan index; $K \leq A$ is the final accepted-plan index.
k_a	Accepted-plan index available at the start of attempt a .
$\tilde{\Pi}_a$	Complete cumulative successor plan proposed by attempt a ; it is not an accepted plan unless its disposition is accepted.
δ_a	Typed attempt disposition in $\{\text{accepted}, \text{rejected}, \text{stop}\}$.
Π_k	Complete cumulative accepted plan k ; Π_0 is the admitted initial fixed or identity plan.

Symbol	Normative meaning (continued)
$\mathbf{u}^{(0)}$	Original admitted aligned raw pair $((\mathbf{x}^A)^\top, (\mathbf{r}^A)^\top)^\top$ used for canonical learning, evaluation, replay, and restart identity.
$\mathbf{u}^{\text{eval},(k)}$	Exploratory paired evaluation product for accepted plan k , normally obtained by applying Π_k to the original admitted pair $\mathbf{u}^{(0)}$.
$\mathcal{G}_{\text{pair}}$	Exact common detector-time/output index grid and representative-occurrence identities, independent of either coordinate's numerical availability.
$\mathcal{A}_{\text{pair}}$	Common pair-action and ordinary-operator support, including masks, segments, guards, state/reset, and transitive ordinary response support.
$\mathcal{A}_x, \mathcal{A}_r$	Coordinate-local numerical availability sets for conditioned x and requested conditioned r on $\mathcal{G}_{\text{pair}}$.
$\mathcal{A}_C$	Joint covariance support, a subset of $\mathcal{A}_x \cap \mathcal{A}_r$ on which the admitted joint statistical model and required covariance components are available.
ρ_{dn}	Representative aligned occurrence (d, Mn) on the raw paired ALIGN parent grid, before replacement and temporal propagation.
$\alpha_{x,d}, \alpha_{r,d}$	Local optical-response coefficients in the paired readout coordinates over a declared signal and support domain.
ϵ_d, ψ_d	Optional optical-leakage ratio $\alpha_{r,d}/\alpha_{x,d}$ and metric-qualified response-direction angle. Their coordinate units, scales, mapping revision, normalization, metric, domain, and comparison compatibility are part of the product; neither parameterization is universal.
$s_{d,\zeta}^x, s_{d,\zeta}^r$	Declared nonzero coordinate-normalization scales used only to place x/r response components in a common metric when an angle is reported.
$a_g(t)$	Declared atmospheric optical-mode template or latent common mode for network g .
$\tau_e, \delta\tau_{de}, \tau_{de}, \Delta\mathbf{b}_{de}$	Network event time, detector-specific timing offset, detector transition time $\tau_{de} = \tau_e + \delta\tau_{de}$, and paired amplitude $(\Delta x_{de}, \Delta r_{de})^\top$; the offset is zero or bounded by a declared coherence tolerance.
$I_h, \mathcal{P}_{d,h}$	Coherent plateau interval h and its application-context-specific admissibility state.
$K_{dn,qj}^{\text{RTC}}$	RTC-local detector-time response of one output cell to one admitted aligned input cell at fixed realized state.
$K_{dn,qj}^{x \leftarrow x, \text{RTC}}, K_{dn,qj}^{r \leftarrow r, \text{RTC}}$	Coordinate-diagonal RTC-local responses for conditioned x and available conditioned r . Both cross-coordinate numerical responses are zero.
$K_{dn,qm}^{\text{end}}$	Optional end-to-end response to one native acquired cell, available only by explicit composition with admitted upstream ALIGN response.
$\mathcal{S}_{dn}$	Exact transitive input support of output (d, n) , including alignment, donors, filters, state, and selection.
$\mathcal{I}_{dn}$	Cause-preserving transitive influence set for output (d, n) .

1.2 Identity and interval conventions

Observation, Tune, array, network/interface, detector occurrence, selected APT row/binding, native row, aligned slot, scan, science interval, context interval, conditioned output slot, stage, and product identities are non-interchangeable. Array identity is one of **a1100**, **a1400**, and **a2000**; array ID, array index, network ID, detector index, and container position do not substitute for one another. Cross-artifact identity requires an explicit occurrence- and product-scoped binding. Equal row numbers do not establish identity.

Primary detector timestreams are samples on rows and detectors on columns unless a product explicitly declares another ordered shape. Every interval is half-open. Science windows, physical scans, processing chunks, filter context, output selections, and learned-plan scan sets retain separate identities. Native, aligned, and output indices never silently renumber one another.

Time is seconds on the admitted ALIGN-assigned grid. The approximately 8-ms rate relationships and the $0.5\times$, $2\times$, and $4\times$ family are conditional software-grid facts, not proof of a detector integration event, absolute epoch, or astrometric correction. Coordinate-dependent Pointing, OOF, and Beammap controls use a declared AltAz tangent plane; Science controls use a declared equatorial J2000 TAN relation. RTC performs no implicit frame conversion.

1.3 Logical and availability notation

$\mathbb{I}[\cdot]$ denotes an indicator, $\mathbb{E}$ expectation, Cov covariance, and $\text{supp}(\cdot)$ operator support. The logical symbols $\wedge$, $\vee$, and $\neg$ have their usual meanings. A quantity decorated available is present with its required identity, domain, and validity; a quantity decorated unavailable is explicitly unavailable. Neither zero nor NaN is an availability encoding.

End of notation; shared definitions follow without a change of authority.

2 Normative definitions

SCI-RTC-DEF-001 – Admitted aligned paired stream.

An immutable observation-scoped pair of **xs** and **rs** arrays. Every admitted x has exactly one r partner with the same observation, Tune, array, network/interface, detector occurrence, native acquisition occurrence, aligned slot, cadence, phase, and mapping lineage. Missing a partner is malformed required input. The paired samples retain independent finiteness, origin/synthesis, support, validity, and uncertainty states.

SCI-RTC-DEF-002 – RTC application context.

The immutable request-side description $\mathcal{R}$ naming requested use, paired input and interval, raw-coordinate sign and reference/baseline, valid domain, named consumers, and external constraints. Historical labels such as Beammap, Pointing, OOF, Science, or diagnostic may be recorded as context but do not partition the admitted RTC operation classes or silently select a policy. Every context consumes paired raw **xs/rs**; when conditioned r is requested it is produced as the paired companion on the selected conditioned- x grid. Only conditioned x is eligible for a later SCI-CAL handoff. A calibrated downstream detector signal is a distinct SCI-CAL product c^{CAL} in the top-of-atmosphere, point-source-equivalent mJy-per-fixed-nominal-beam convention; it is not another meaning of **xs**.

SCI-RTC-DEF-003 – Exact detector occurrence.

An observation-local layered binding containing observation, Tune, network/interface, network-local tone or admitted row slot, selected APT identity/digest, and row association. A transient column number is only a locator.

SCI-RTC-DEF-004 – Downstream CAL handoff.

An immutable RTC-to-SCI-CAL boundary carrying the raw conditioned parent, detector binding, complete RTC plan/response/support, selected **flxscale** authority, target-atmosphere lineage, validity, uncertainty availability, and provenance required by SCI-CAL. Absolute **flxscale** and target atmosphere are applied only after RTC; RTC does not derive, apply, repair, or promote the calibrated product. Because target-atmosphere correction is sample dependent and follows temporal filtering, a handoff claim also requires complete response accounting over RTC support or a preregistered owner-approved noncommutation bound; output-time correction is not an automatic inverse of a filtered atmosphere operator.

SCI-RTC-DEF-005 – Compatible flxscale pair.

Two finite factors ϕ_q and ϕ_d that are valid for the exact donor and target detector occurrences under the same multiplicative calibration convention, quantity domain, reference plane, and applicable support, with $\phi_d \neq 0$. Compatibility is not established by matching row numbers or finite values alone.

SCI-RTC-DEF-006 – Selected conditioning policy.

A versioned, admitted specification selected within a resolved RTC plan from operation classes available to every application context. It covers spike-candidate detection and change-point masking/downweighting, donor eligibility, donor scoring and ties, post-segmentation x replacement and fallback, source protection, filter stages and order, exact coefficients and normalization, numeric precision, state/reset, boundaries/edges, non-finite handling, sampling mode, phase-zero factor, flags, influence aggregation, diagnostics, and failure behavior. Availability or admission does not mean enabled, selected, numerically resolved, or scientifically qualified. It is selected authority, not an implementation default.

SCI-RTC-DEF-007 – One-way context–plan–record lifecycle.

Three distinct objects are preserved in one-way order. The *RTC application context* preserves requested use, input, consumers, and constraints. The *RTC resolved plan* is the immutable selected numerical plan admitted after every required predicate passes, including any learned evidence and observation binding. The *RTC realized record* states the exact operations, intervals, state, exceptions, outputs, and completion that actually occurred. More detailed requested, effective, observation-resolved, learned, resolved, and realized substates may be retained, but a later object or substate never rewrites an earlier one.

SCI-RTC-DEF-008 – Realized conditioning operator.

The chronological composition of paired admission and evidence selection, level-shift transition masking and

plateau segmentation, then every admitted x -domain replacement, mask, FIR/IIR/notch, boundary, reset state, non-finite, and phase-zero sampling action, including all affine state terms and exception intervals. Raw r may affect evidence and the resolved pair state; every ordinary canonical stage then acts through the same L_{Π} on both coordinates, without copying any x -only donor exception into r . Annotations that affect values, support, influence, or availability are part of the operator. Detection and affected-support decisions are pair-coherent; coordinate-specific evidence and correction amplitudes remain distinct.

SCI-RTC-DEF-009 – Phase-zero sampling.

Point selection of the pre-sampling paired stream at input indices $j = Mn$ for one integer factor $M \geq 1$, with conditioned r occupying exactly the conditioned- x grid when requested. It is not an arithmetic mean, block aggregate, or implicit anti-alias filter. Its representative occurrence is exactly (d, Mn) on the raw paired aligned parent before replacement and filtering; that identity is not a largest-coefficient claim.

SCI-RTC-DEF-010 – Complete realized response.

An exact local or factorized representation that includes every enabled response-changing stage, detector mixing, mask, state, edge rule, exception, and sampling phase on its declared support. Conditioned x and available conditioned r use identical ordinary diagonal response blocks; both cross-coordinate numerical responses are zero. The RTC-local response is with respect to the admitted aligned pair. An optional end-to-end response to the native acquisition grid is a distinct composition with admitted upstream ALIGN response and lineage. If either representation is not truthfully available, its response state is unavailable; a partial kernel is not complete response.

SCI-RTC-DEF-011 – Representative source occurrence.

For output (d, n) , the raw aligned paired occurrence (d, Mn) before replacement and temporal propagation. It may be original, ALIGN-synthesized, or subsequently RTC-replaced. This identity, rather than the largest realized coefficient, governs independent-exposure exclusion.

SCI-RTC-DEF-012 – Transitive influence.

The cause-preserving closure from every original, synthesized, replaced, masked, edge, state, and non-finite source cell through all nonzero realized alignment, donor, filter, state, and sampling dependencies to an output. Influence is broader than the selected point and may cross detector columns.

SCI-RTC-DEF-013 – Direct exclusion and consumer eligibility.

An output whose representative occurrence of SCI-RTC-DEF-011 is ALIGN-synthesized or RTC-replaced is universally excluded as an independent detector measurement, even when finite. Synthesis or replacement elsewhere in its transitive support remains a mandatory cause-preserving influence input, not a universal RTC eligibility decision. Each downstream consumer owns the declared policy that maps that influence, response, and support to its own eligibility.

SCI-RTC-DEF-014 – Coordinate-dependent control.

A source mask or response operation that consumes an AST coordinate identity, frame, topology, detector binding, validity, uncertainty where required, and support. Invalid or unavailable coordinates are not outside-source values.

SCI-RTC-DEF-015 – Fixed sampling mode.

A request whose admitted factor, realizable prefilter, rate, phase-zero rule, and downstream compatibility are resolved without learning them from the observation.

SCI-RTC-DEF-016 – Learned sampling mode.

An optional request that evaluates an admitted finite candidate set and resolves one immutable observation-wide cadence/filter plan before apply. Its admission uses the smallest admitted beam, maximum valid in-scan speed, analytical beam-times-realized-filter response, exact phase-zero alias accounting, sampling sufficiency, filter realizability, and downstream compatibility. Noise-aware, per-array, per-scan, heterogeneous-cadence, and continuous adaptation are outside v0.1.

SCI-RTC-DEF-017 – Conditional statistical covariance.

The covariance obtained from an admitted joint x/r input covariance through the fixed identical paired operator while holding selection and nuisance state fixed. It preserves available cross-coordinate covariance even though numerical $x \leftrightarrow r$ response is zero. It is not total uncertainty and does not imply a diagonal, white, stationary, or independent noise model. Every covariance or uncertainty claim, whether conditional, component-limited, or total, explicitly names the components and correlations it includes and excludes; unknown items remain unavailable rather than zero.

SCI-RTC-DEF-018 – Atomic RTC bundle.

One consumer-neutral completion unit containing conditioned raw x TOD; when requested, its grid-aligned conditioned raw r companion and exact local or global availability; the immutable paired **rs** parent and pairing/mapping identity; every **rs**-derived diagnostic, selector, segmentation, and resolved-plan use; ordered detector/sample/scan/support and parent identity; complete response or typed unavailability; cause-preserving flags and influence; separated validity and eligibility inputs; conditional uncertainty/covariance status; one-way state

and provenance; required common diagnostics and any separately requested diagnostic detail; downstream CAL handoff state; and completion/failure state. Conditioned r is optional product content under the same context, plan, and realized record; it is not a separately optimized pipeline. TOD alone is not this bundle.

SCI-RTC-DEF-019 – Scientifically named diagnostic.

A quantity with an explicit estimator, unit, support, selection/mask, validity, uncertainty/availability, and policy role. It is either inert, advisory, or an input to an explicitly selected policy; those roles are not interchangeable.

SCI-RTC-DEF-020 – Claim layer.

One of algebraic contract correctness, implementation conformity, representation fidelity, observational performance, science-impact qualification, or production readiness. Evidence at one layer does not establish another.

SCI-RTC-DEF-021 – Declared learning population.

The exact training interval, designated same-observation interval, scan collection, observation aggregate, or other named population used to infer an RTC quantity, including identity, support, selection, masks, exclusions, signal role, compatibility domain, and completeness. A population is not defined by whatever samples happen to be available during execution.

SCI-RTC-DEF-022 – Learned evidence.

The output of named diagnostics applied to a declared learning population, including candidate parameters, estimator, unit and convention, support, quality, uncertainty, validity, and failure state. Learned evidence proposes or constrains a plan; it does not authorize apply.

SCI-RTC-DEF-023 – Resolved RTC plan.

One immutable observation- and application-context-bound specification admitted only after all required scientific predicates pass. It contains operation sequence, family, exact coefficients or identity, rates and conventions, order/taps, bands and tolerances, notch state, applicability, masks, donor policy, edges, guards, resets, decimation and timing, expected response/support, uncertainty, validity, and downstream compatibility.

SCI-RTC-DEF-024 – Apply state.

Execution of exactly one resolved plan on compatible admitted input. Apply records actual intervals, exceptions, resets, outputs, and completion but does not relearn, retune, change policy, or rewrite the resolved plan.

SCI-RTC-DEF-025 – Online-adaptive estimator.

A separate time-varying, data-dependent estimator whose coefficients or other operator parameters evolve during apply under a declared update rule. Its learning support, stability, response, uncertainty, guard, reset, and restart semantics are not those of learned-then-fixed filtering. It is unavailable in v0.1 absent successor authority.

SCI-RTC-DEF-026 – Scientific filter design specification.

The physical purpose, source/contaminant model, admitted scan/beam/source domain, pass/transition/stop or notch constraints, phase and registration, alias bound, support/guard, numeric precision, realizability, validation, and failure criteria that justify a filter family and its numerical parameters. An operation label or coefficient array alone is not this specification.

SCI-RTC-DEF-027 – Bounded iterative notch-plan refinement.

A finite owner-bounded sequence of refinement attempts. Each attempt learns from a declared population and the current accepted-plan paired evaluation product derived from the original admitted pair $\mathbf{u}^{(0)}$, resolves a complete cumulative proposal and typed disposition, and advances the accepted-plan index only when that proposal is accepted. Apply of every accepted plan is immutable; rejected or stop attempts do not manufacture a new accepted plan or evaluation product. This is not the online-adaptive estimator of SCI-RTC-DEF-025.

SCI-RTC-DEF-028 – Complete cumulative successor proposal.

The full plan proposed by attempt a and accepted as Π_{k+1} only after its admission predicates pass, containing every retained, modified, removed, and rejected notch disposition; every broad-band stage; exact order, coefficients, precision, state, masks, edges, guards, intervals, decimation, timing, cumulative response/support, uncertainty, validity, and downstream compatibility. “Add this notch to the prior output” is not a complete successor plan. Its canonical evaluation and final replay parent is the original admitted pair, not an x -only projection or accumulated exploratory output.

SCI-RTC-DEF-029 – Final accepted plan and termination disposition.

The last complete plan whose admission predicates passed, together with one typed finite stop cause. No-successor, residual-satisfied, stable-plan, budget/stability/support/compatibility rejection, insufficient evidence, maximum-attempt, oscillating-plan, nonconvergent-parameter, and owner-cancelled states remain distinct. Maximum-attempt termination is not convergence.

SCI-RTC-DEF-030 – Paired raw x, r detector coordinates.

The two raw coordinates $\mathbf{u}_d = (x_d, r_d)^T$ produced for one exact detector occurrence by an admitted Tune-dependent readout mapping. The x coordinate carries the ordinary total-intensity analysis path; the nominally orthogonal r coordinate carries readout, leakage, artifact, and Tune-health information. Pairing does not assert equal scaling,

response, validity, or uncertainty.

SCI-RTC-DEF-031 – Tune-dependent $I, Q \rightarrow x, r$ mapping authority.

The upstream authority identified by ζ that owns the microwave transmission coordinates, transform, reference point, normalization, sign, local Jacobian or nonlinear equivalent, validity/linearity domain, epoch, and uncertainty. RTC consumes its identity and admitted output but does not produce or repair the mapping.

SCI-RTC-DEF-032 – Optical leakage into the paired r coordinate.

A nonzero admitted optical-response projection in nominally orthogonal r . It is a readout-coordinate alignment or optical-leakage diagnostic, not automatically a literal Tune-angle error. A scalar ratio or angle tests only its declared optical projection and does not prove the complete transform. Every product retains the x/r coordinate units or scales, mapping revision, normalization and metric, Tune and validity domain, whether its ratio is dimensionless or carries a coordinate-unit ratio, and the compatibility rule for comparison across detectors, Tunes, loading states, epochs, or mapping revisions. An angle is available only after a declared metric makes the two axes commensurate. Otherwise it is coordinate bookkeeping, not a comparable physical readout angle.

SCI-RTC-DEF-033 – Atmospheric optical-leakage diagnostic.

A scientifically named estimate of paired x/r response to a declared network atmospheric optical-mode template, with exact membership, self-inclusion rule, time/frequency support, preprocessing, estimator, response, covariance, uncertainty, coherence, Tune/loading state, and scalar, frequency-dependent, or unavailable disposition. The estimator shall address shared-data dependence, noise in both coordinates, and detector self-inclusion; measured correlation or direct noisy r -on- x regression is not by itself an unbiased leakage coefficient. This diagnostic does not authorize numerical template subtraction from x .

SCI-RTC-DEF-034 – Bright-source optical-leakage diagnostic.

A separately parented estimate of paired x/r response to a declared localized source template and support, often requested by Beammap, Pointing, or OOF application contexts. It neither inherits nor silently replaces an atmosphere-derived diagnostic.

SCI-RTC-DEF-035 – Network-coherent detector-dependent level shift.

An unexpected persistent additive readout-baseline change having a common or near-common network event identity, detector transition support constrained by a declared coherence tolerance, and independently estimated detector/channel amplitudes ($\Delta x, \Delta r$). On each stable x plateau the v0.1 model is signal plus an additive baseline; it contains no multiplicative gain, responsivity, or more general response-change degree of freedom. Simultaneity, an r signature, or its absence does not by itself identify physical cause.

SCI-RTC-DEF-036 – Transition interval and guarded change-point support.

The finite physical-time interval between stable pre- and post-event plateaus, together with its timing uncertainty, affected detector population, and any downstream operator-specific guards. Its sample support is derived from the applicable timing vector, never from a fixed sample count or scan-relative fraction. The physical transition support is explicitly flagged, excluded from both plateau estimators, and remains unmodeled even when a valid plateau offset correction is applied. Propagated influence remains distinguishable from this underlying physical support. Reset is the ordinary segmentation behavior; carry requires separately authorized continuity.

SCI-RTC-DEF-037 – Stable plateau and application-context admissibility.

A half-open interval outside resolved transition support that is consistent with its declared signal-plus-additive-baseline model, accounting for atmosphere, sources, noise, slow variation, Tune state, valid support, and absence of unresolved additional shifts. Suitability is application-context specific and depends on the required estimator support, not sample count alone.

SCI-RTC-DEF-038 – Separately authorized response-changing correction.

Any estimator that numerically mixes x and r , subtracts or rotates one into the other, or assigns a multiplicative gain, responsivity, or more general response change to a level-shift event. It requires separate purpose, coupling and learning models, complete response and covariance, astronomical transfer, uncertainty, validation, restart, and provenance authority; none is part of the v0.1 additive level-shift model. A selected additive translation between sufficiently supported stable x plateaus is governed instead by Equations (SCI-RTC-EQ-033)–(SCI-RTC-EQ-034) and REQ-101.

SCI-RTC-DEF-039 – Conditioned- r paired companion.

The optional raw-coordinate r product produced under the same RTC application context, immutable resolved plan, realized record, selected grid, temporal support, source protection, and pair-level pathology decisions as conditioned x . Ordinary shared filtering, masking, state, and sampling act coordinate-diagonally. Existing x repair does not imply r reconstruction: where no scientifically justified r -domain correction exists, the paired r location and response remain typed invalid or unavailable with exact cause. The companion preserves raw- r parentage, mapping, units, sign, reference, optical-leakage lineage, coordinate-specific evidence, response, uncertainty status, and provenance. It does not enter SCI-CAL, numerically mix into conditioned x , or authorize numerical

$x \leftrightarrow r$ mixing; its evidence may nevertheless affect the resolved pair plan under equations (SCI-RTC-EQ-037)–(SCI-RTC-EQ-039).

SCI-RTC-DEF-040 – Canonical conditioned pair.

The jointly requested RTC product formed from one exact admitted raw x/r occurrence pair, one application context, one immutable resolved pair plan, one realized record, and one selected output grid. On ordinary common available support, both coordinates undergo the identical operator of equation (SCI-RTC-EQ-040). “Conditioned pair” is not a permitted product claim when conditioned r was not requested or is globally unavailable; the product is then conditioned x with raw- r parentage and the exact conditioned- r request/availability disposition.

SCI-RTC-DEF-041 – Coordinate-specific evidence.

A learned record E_x or E_r whose direct numerical observation originates in exactly one member of the raw pair. It preserves detector and occurrence identity, estimator, coordinate of origin, physical and sample support, source-protection and Tune/mapping state, uncertainty, quality, causal class, direct status, and source or frequency domain where applicable. Pair action does not convert it into direct evidence in the other coordinate.

SCI-RTC-DEF-042 – Joint paired evidence.

A learned record E_{xr} whose estimator and declared causal meaning depend on the joint behavior of the exact paired coordinates. It carries the same identity, support, protection, mapping, uncertainty, quality, and domain facts as coordinate-specific evidence and is not reducible to two generic flags.

SCI-RTC-DEF-043 – Hard pair-pathology class.

An owner-designated causal class whose accepted direct or joint evidence is sufficient under its resolved policy to require pair-level action. Accepted spikes, level shifts, resets or discontinuities, structurally invalid mapped pairs, declared non-finite or out-of-domain events, and later expressly approved classes may qualify. Weak lines, PSD shape, variance, leakage, coherence, population outliers, Tune-health metrics, and source-overlapping features remain diagnostic or candidate evidence until their stronger owner-approved predicates pass.

SCI-RTC-DEF-044 – Pair-level action support.

The immutable mask, physical/sample interval, guards, segments, reset boundaries, and transitive operator influence on which one resolved pair plan acts because at least one accepted hard cause requires action. Its union does not merge or erase the originating evidence records or their causal classes.

SCI-RTC-DEF-045 – Inferred pair contamination.

The typed consequence assigned to the paired coordinate when accepted direct evidence in one coordinate or accepted joint evidence establishes a hard pair event. It identifies direct- x , direct- r , joint, inferred- x -from- r , or inferred- r -from- x status and never asserts a numerical detection that was not made.

SCI-RTC-DEF-046 – Common identical paired operator.

The canonical ordinary operator $I_2 \otimes L_\Pi$, with exactly the same masks, pair-action support, segments, transitions, guards, coefficients, state, initialization/reset, filter definitions, order, precision, phase/delay, boundaries, down-sampling, selected times, representative occurrences, cardinality, and output grid on both coordinates. Its two cross-coordinate numerical branches are zero.

SCI-RTC-DEF-047 – Coordinate-specific affine correction.

An admitted additive baseline translation b^x or b^r learned for one coordinate on stable plateau support. A shared event has one transition, guards, segmentation, and reset, while the two amplitudes, uncertainties, quality states, availability, and numerical correction actions remain separate. Neither amplitude is copied, inferred, or numerically mixed from the other.

SCI-RTC-DEF-048 – x -only donor-recovery exception.

The existing coordinate-specific RTC repair in which justified donor information may replace an accepted target x occurrence. It is not an ordinary common paired operator, creates no donor-derived r value, and makes conditioned r and its response unavailable over the full causal influence of the repair while retaining the common grid and every raw-pair cause.

SCI-RTC-DEF-049 – Coordinate-local unavailable influence.

The exact selected-grid support on which one conditioned coordinate, its response, or its covariance component cannot be scientifically produced, including the complete downstream transitive influence of an x -only donor repair or a required but unavailable coordinate-specific affine correction. Finite filters, stateful filters, and phase-zero sampling propagate the cause through their exact realized support; an IIR influence may continue to the next reset. Locations remain in place with typed causes, and the other coordinate may remain available.

SCI-RTC-DEF-050 – Separately optimized r diagnostic descendant.

Any future product that uses an r -specific temporal, sampling, state, mask, or response operator differing from L_Π . It requires a new name, identity, purpose, response, covariance, uncertainty, provenance, and owner authority and is not the canonical conditioned- r companion.

SCI-RTC-DEF-051 – Cause-preserving pair evidence/action record.

The ordered record linking every E_x , E_r , and E_{xr} item to its accepted/rejected/candidate disposition, direct or inferred status, resolved pair action, exact support and transitive influence, coordinate-local modification and availability, operator identity, uncertainty, and downstream handoff. It allows reconstruction of why pair action occurred without turning a unioned support into an undifferentiated flag.

SCI-RTC-DEF-052 – Pair support and availability partition.

Four noninterchangeable states: (1) the common index grid $\mathcal{G}_{\text{pair}}$; (2) common pair-action and ordinary-operator support $\mathcal{A}_{\text{pair}}$; (3) coordinate-local numerical availability $\mathcal{A}_x$ and $\mathcal{A}_r$; and (4) joint covariance support $\mathcal{A}_C \subseteq \mathcal{A}_x \cap \mathcal{A}_r$ where both conditioned values and the required admitted statistical model are available. Common identity or action support never asserts identical numerical availability.

3 Normative equations and identities

The equations in this section are the sole displayed mathematical authority of both views. They describe an implementation-independent, state-conditioned operator. They neither select numerical policy nor assert implementation conformity.

3.1 Raw RTC domain, donor transfer, and CAL handoff

For an exact detector occurrence i under the admitted multiplicative **flxscale** convention, define the detector-static comparison coordinate

$$\chi_i = \phi_i x_i, \quad \phi_i = \text{flxscale}_i. \quad (\text{SCI-RTC-EQ-001})$$

Equation (SCI-RTC-EQ-001) neither produces a calibrated RTC product nor assigns a role to legacy **responsivity**; target-atmosphere correction is absent. When q is a raw donor and d is a raw target, the unique multiplicative transfer that represents donor-calibrated amplitude in the target raw convention is

$$x_{d \leftarrow q}^R = a_{d \leftarrow q} x_q, \quad a_{d \leftarrow q} = \frac{\phi_q}{\phi_d}, \quad \chi_d^R = \phi_d x_{d \leftarrow q}^R = \phi_q x_q = \chi_q. \quad (\text{SCI-RTC-EQ-002})$$

This equation is defined only for a compatible **flxscale** pair as defined above. Otherwise this transfer is unavailable. An additive calibration convention, a different unit/reference domain, or an alternative donor authority requires its own complete affine equivalence and is not silently folded into equation (SCI-RTC-EQ-002).

Let $\Phi = \text{diag}(\phi_d)$ be the selected detector-static factor operator and $C_{\kappa, \mathcal{R}}^{\text{atm}}$ the target-atmosphere operator owned by SCI-CAL. The distinct downstream calibration operator is

$$C_{\kappa, \mathcal{R}}^{\text{CAL}} = \Phi C_{\kappa, \mathcal{R}}^{\text{atm}}, \quad \mathbf{c}_{\mathcal{R}}^{\text{CAL}} = C_{\kappa, \mathcal{R}}^{\text{CAL}} \mathbf{y}_{\mathcal{R}}^{x, \text{RTC}}. \quad (\text{SCI-RTC-EQ-003})$$

Equation (SCI-RTC-EQ-003) is an SCI-CAL handoff identity; its output is not **xs** and it is not an RTC stage. The selected cross-package boundary is

$$\begin{aligned} \begin{bmatrix} \mathbf{x}^A \\ \mathbf{r}^A \end{bmatrix} &= A_\alpha \begin{bmatrix} \mathbf{x}^{\text{acq}} \\ \mathbf{r}^{\text{acq}} \end{bmatrix} + \mathbf{a}_\alpha & (\text{upstream ALIGN relation}), \\ (\mathbf{x}^A, \mathbf{r}^A) &\xrightarrow{\text{RTC pair admission, spike evidence, shift resolve}} \Omega_{xr}, \\ \mathbf{x}^A &\xrightarrow{B_\omega^x, F_1 \dots F_K, D_{M,0}} \mathbf{y}_{\mathcal{R}}^{x, \text{RTC}} \xrightarrow{C_{\kappa, \mathcal{R}}^{\text{CAL}}} \mathbf{c}_{\mathcal{R}}^{\text{CAL}}, \\ \mathbf{r}^A &\xrightarrow{\text{same applicable } F_1 \dots F_K, D_{M,0}} \mathbf{y}_{\mathcal{R}}^{r, \text{RTC}} \text{ when requested,} \\ \mathbf{r}^{A, \text{parent}}, \mathbf{y}_{\mathcal{R}}^{r, \text{RTC}}, \mathcal{D}_{\mathcal{R}}^r &\hookrightarrow \mathcal{O}_{\text{RTC}, \mathcal{R}}. \end{aligned} \quad (\text{SCI-RTC-EQ-004})$$

RTC shall not move the downstream CAL arrow inside replacement, temporal filtering, or phase-zero sampling. Only the x member is handed to SCI-CAL. The paired raw r parent and every causal diagnostic/selecter remain in the bundle. When conditioned r is requested, its ordinary shared RTC filtering and phase-zero selection are coordinate-diagonal; no x repair or replacement is silently applied to it. Neither raw nor conditioned r is calibrated, substituted for x , or interpreted as Stokes data. The raw donor ratio in equation (SCI-RTC-EQ-002) is x -domain

convention transfer within B_ω^x , not absolute calibration. Atmospheric templates are diagnostic evidence only; equation (SCI-RTC-EQ-004) contains no atmospheric-removal operator. The first line is retained upstream lineage, not an RTC stage. RTC begins at the admitted aligned pair; A_α is composed again only when a separately identified end-to-end response to native acquisition coordinates is requested.

3.2 Realized conditioning operator

Conditioned on complete realized state Ω —which may depend on paired x/r evidence and selectors—the required ordered raw x operator is

$$\mathbf{y}_{\mathcal{R}}^{x,\text{RTC}} = D_{M,0} F_{K,\theta_K} \cdots F_{1,\theta_1} B_{\omega,\mathcal{R}}^x \mathbf{x}^A + \mathbf{b}_\Omega^x. \quad (\text{SCI-RTC-EQ-005})$$

Chronological order is read from right to left. The affine terms include only declared replacement, filter-state, boundary, or other authorized contributions. Masks, coefficients, finite handling, context, exceptions, and state are components of Ω , not annotations outside the operator. Conditional on them,

$$\begin{aligned} \mathbf{y}_{\mathcal{R}}^{x,\text{RTC}} &= L_\Omega^x \mathbf{x}^A + \mathbf{b}_\Omega^x, & L_\Omega^x &= D_{M,0} F_K \cdots F_1 B_{\omega,\mathcal{R}}^x, \\ K_\Omega^{x \leftarrow x, \text{RTC}} &= \left. \frac{\partial \mathbf{y}^x}{\partial \mathbf{x}^A} \right|_\Omega = L_\Omega^x, & K_\Omega^{x \leftarrow r, \text{RTC}} &= \left. \frac{\partial \mathbf{y}^x}{\partial \mathbf{r}^A} \right|_\Omega = 0, \\ J_\Omega^{\text{end}} &= \begin{bmatrix} L_\Omega^x & 0 \end{bmatrix} A_\alpha & & \text{when the upstream ALIGN response is admitted.} \end{aligned} \quad (\text{SCI-RTC-EQ-006})$$

This is a conditional affine identity, not an unconditional linear, stationary, unbiased, or Gaussian claim.

A realized replacement at target cell (d, j) may be written

$$x_{dj}^R = (1 - p_{dj}) x_{dj} + p_{dj} \left[\sum_{(q,\ell) \in \mathcal{Q}_{dj}} w_{dj,q\ell} a_{d \leftarrow q,j\ell} x_{q\ell} + \beta_{dj} \right], \quad \sum_{q,\ell} w_{dj,q\ell} = 1 \text{ when normalization is claimed.} \quad (\text{SCI-RTC-EQ-007})$$

This replacement is explicitly x -only. The scale a is domain-qualified: equation (SCI-RTC-EQ-002) is the authorized raw `flxscale` route. Any other transfer carries separate identity and validity and does not import downstream calibration into RTC.

For a time-invariant FIR stage,

$$v_{d,j} = \sum_{\ell \in \mathcal{L}_k} h_{d,\ell}^{(k)} u_{d,j-\ell}, \quad H_d^{(k)}(e^{i\omega}) = \sum_{\ell \in \mathcal{L}_k} h_{d,\ell}^{(k)} e^{-i\omega\ell}, \quad H_d^{(k)}(1) = \sum_{\ell} h_{d,\ell}^{(k)}. \quad (\text{SCI-RTC-EQ-008})$$

For a causal IIR or notch stage, a state-complete form is

$$\mathbf{s}_{j+1} = A_k \mathbf{s}_j + B_k u_j, \quad v_j = C_k \mathbf{s}_j + D_k u_j, \quad \mathbf{s}_{j_0} = \mathbf{s}_{0,k}. \quad (\text{SCI-RTC-EQ-009})$$

An exactly equivalent recurrence is admissible only with its coefficient convention, precision, state, and boundary identity. A causal-stability claim requires every realized pole strictly inside the unit circle.

One possible selected mask operator is normalized masked FIR convolution,

$$v_j = \frac{\sum_{\ell} h_{\ell} m_{j-\ell} u_{j-\ell}}{\sum_{\ell} h_{\ell} m_{j-\ell}}, \quad \left| \sum_{\ell} h_{\ell} m_{j-\ell} \right| > \epsilon_m, \quad (\text{SCI-RTC-EQ-010})$$

but equation (SCI-RTC-EQ-010) is not a selected default. Segmenting, zero filling, restoring a mask after filtering, holding state, and renormalizing are distinct operators.

3.3 Phase-zero sampling and response

For a coherent pre-sampling segment of length N , v0.1 selects only

$$\begin{aligned} y_{d,n}^{x,\text{RTC}} &= v_{d,Mn}^{x,\text{preD}}, & y_{d,n}^{r,\text{RTC}} &= v_{d,Mn}^{r,\text{preD}} \text{ when available,} \\ 0 \leq Mn < N, & f_{\text{out}} = \frac{f_{\text{in}}}{M}, & N_y &= \begin{cases} 0, & N = 0, \\ 1 + \left\lfloor \frac{N-1}{M} \right\rfloor, & N > 0. \end{cases} \end{aligned} \quad (\text{SCI-RTC-EQ-011})$$

The conditioned- r companion has the same cardinality and selected slots as conditioned x when requested. A location at which no scientifically justified conditioned- r value exists remains present and typed invalid or unavailable; it is not dropped, reindexed, interpolated, or filled. The representative assigned-grid time is the selected point time $t_n^{\text{out}} = t_{Mn}$; the scientific support is nevertheless the complete RTC-local transitive support $\mathcal{S}_{dn}^{\text{RTC}}$ of all preceding stages. A selected edge/output policy may reject candidate points, but shall not relabel a different phase or silently change equation (SCI-RTC-EQ-011). The representative source occurrence is exactly $\rho_{dn} = (d, Mn)$ in the raw aligned paired parent. It is not the largest response coefficient, a support centroid, a donor occurrence, or a synthetic sample.

For output (d, n) and admitted aligned input (q, j) , the exact RTC-local response is

$$K_{dn,qj}^{\text{RTC}} = \frac{\partial y_{dn}}{\partial x_{qj}^A} \quad \text{at fixed realized state } \Omega. \quad (\text{SCI-RTC-EQ-012})$$

At a selection boundary the derivative may be discontinuous or undefined; that state shall be represented rather than replaced by an LTI claim. An end-to-end kernel to native acquisition cells is the distinct optional composition already defined in equation (SCI-RTC-EQ-006).

In the fixed, detector-separable, translation-invariant, single-rate interior case only,

$$H_d^{\text{comb}}(e^{i\omega}) = \prod_{k=1}^K H_d^{(k)}(e^{i\omega}), \quad \tau_g(\omega) = -\Delta t \frac{d}{d\omega} \text{unwrap}[\arg H_d^{\text{comb}}(e^{i\omega})]. \quad (\text{SCI-RTC-EQ-013})$$

Amplitude is $|H|$, power response is $|H|^2$, and phase is $\arg H$ under the recorded Fourier convention. Group delay is unavailable at response zeros unless a bounded convention is supplied.

For an LTI prefilter followed by phase-zero selection, the exact multirate identity is

$$Y(e^{i\Omega_o}) = \frac{1}{M} \sum_{p=0}^{M-1} H(e^{i(\Omega_o+2\pi p)/M}) X(e^{i(\Omega_o+2\pi p)/M}), \quad (\text{SCI-RTC-EQ-014})$$

under the declared Fourier/sign convention. Every folded image contributes; point selection itself is not anti-alias filtering.

For an FIR output, direct support starts with

$$\mathcal{S}_{dn}^{(1)} = \{Mn - \ell : \ell \in \mathcal{L}\}, \quad \mathcal{S}_{dn}^{\text{RTC}} = \text{closure}_{B,F,\text{state}}(\mathcal{S}_{dn}^{(1)}). \quad (\text{SCI-RTC-EQ-015})$$

This is RTC-local support on the aligned grid. Optional end-to-end support is its explicit upstream expansion through admitted A_α support. IIR support extends to the declared state/reset or context boundary and may be mathematically infinite. Any finite guard is an approximation with a stated residual-tail criterion.

3.4 Moments, influence, and learned-plan selection

For admitted paired input $\mathbf{u}^A = (\mathbf{x}^A, \mathbf{r}^A)^\top$ with mean $\boldsymbol{\mu}_u$ and joint covariance $\Sigma_u = \begin{bmatrix} \Sigma_{xx} & \Sigma_{xr} \\ \Sigma_{rx} & \Sigma_{rr} \end{bmatrix}$, conditional on fixed Ω independent of the modeled random noise, define $J_{\text{num},\Omega} = \begin{bmatrix} L_\Omega^x & 0 \end{bmatrix}$ for conditioned x alone. Then

$$\mathbb{E}[\mathbf{y} \mid \Omega] = J_{\text{num},\Omega} \boldsymbol{\mu}_u + \mathbf{b}_\Omega^x, \quad (\text{SCI-RTC-EQ-016a})$$

$$\Sigma_{y|\Omega}^{\text{stat}} = J_{\text{num},\Omega} \Sigma_u J_{\text{num},\Omega}^\top = L_\Omega^x \Sigma_{xx} (L_\Omega^x)^\top. \quad (\text{SCI-RTC-EQ-016b})$$

The cross-coordinate blocks do not enter fixed-state numerical propagation of conditioned x alone. They remain required for a joint selector, estimator, learned parameter, or diagnostic because $\Omega = \Omega(\mathbf{x}^A, \mathbf{r}^A)$. If selectors depend on the random data, unconditional covariance obeys

$$\text{Cov}(\mathbf{y}) = \mathbb{E}_\Omega[\text{Cov}(\mathbf{y} \mid \Omega)] + \text{Cov}_\Omega(\mathbb{E}[\mathbf{y} \mid \Omega]). \quad (\text{SCI-RTC-EQ-017})$$

The second term is not zero by default. With named nuisance parameters $\boldsymbol{\eta}$ and an adequate local linearization,

$$\Sigma_y^{\text{sys}} \approx J_\eta \Sigma_\eta J_\eta^\top, \quad J_\eta = \left. \frac{\partial \mathbb{E}[\mathbf{y} \mid \Omega, \boldsymbol{\eta}]}{\partial \boldsymbol{\eta}} \right|_{\boldsymbol{\eta}_0}, \quad (\text{SCI-RTC-EQ-018})$$

Claims may be conditional, component-limited, or total. Each claim shall name the components and correlations included and excluded. The following is a possible complete decomposition only when every applicable term and cross term is supplied; “total” is reserved for that declared complete coverage:

$$\Sigma_y^{\text{total}} = \Sigma_y^{\text{stat}} + \Sigma_y^{\text{sys}} + \Sigma_y^{\text{selection}} + \Sigma_y^{\text{model}} + \Sigma_y^{\text{cross}}. \quad (\text{SCI-RTC-EQ-019})$$

Unknown or omitted components and correlations are unavailable, not zero. A truthfully labeled partial or conditional claim does not require this maximal decomposition.

Define cause-preserving influence, direct exclusion, and consumer eligibility by

$$\mathcal{I}_{dn} = \text{closure}_{K, \text{generative links}} \{\text{typed source causes on } \mathcal{S}_{dn}\}, \quad (\text{SCI-RTC-EQ-020a})$$

$$E_{dn}^{\text{RTC, direct}} = V_d^{\text{det}} \wedge V_{dn}^{\text{num}} \wedge V_{dn}^{\text{support}} \wedge V_{dn}^{\text{response}} \wedge \neg D_{dn}^{\text{ALIGN_synth}} \wedge \neg D_{dn}^{\text{RTC_replace}},$$

$$E_{dn}^{(c)} = E_{dn}^{\text{RTC, direct}} \wedge P_c(\mathcal{I}_{dn}, \mathcal{S}_{dn}, K_{dn}, \dots) \quad (\text{SCI-RTC-EQ-020b})$$

D_{dn} identifies whether the exact representative occurrence ρ_{dn} was synthesized or replaced before subsequent filtering and selection. P_c is the declared policy of consumer c for noncenter transitive influence. RTC preserves every cause and numerical dependency but does not replace P_c with a universal transitive-ineligibility decision.

For learned mode, let $\mathcal{M}$ be the admitted finite integer candidate set and $\mathcal{F}(M)$ the conjunction of all owner-resolved astronomical transfer, exact phase-zero alias, sampling, realized-filter realizability, and downstream-compatibility constraints, evaluated with the smallest admitted beam and maximum valid in-scan speed. Then

$$\mathcal{M}_{\text{safe}} = \{M \in \mathcal{M} : \mathcal{F}(M) = \text{true}\}, \quad M_{\star} = \max \mathcal{M}_{\text{safe}}. \quad (\text{SCI-RTC-EQ-021})$$

Equation (SCI-RTC-EQ-021) is defined only after every numerical tolerance, support rule, fallback, and compatibility predicate named in the owner ledger is resolved. Percentile speed is diagnostic, not safety authority.

Finally, the consumer-neutral atomic product for application context $\mathcal{R}$ is

$$\mathcal{O}_{\text{RTC}, \mathcal{R}} = (\mathbf{y}^x, \mathbf{y}^{r, \text{opt}}, \mathbf{r}^{A, \text{parent}}, \mathcal{J}^{xr}, \mathcal{K}^x, \mathcal{K}^{r, \text{opt}}, \mathcal{S}, \mathcal{I}, \mathcal{F}, \mathcal{V}, \mathcal{U}, \mathcal{P}, \mathcal{D}, \mathcal{C}), \quad (\text{SCI-RTC-EQ-022})$$

where $\mathcal{J}^{xr}$ is exact ordered pair identity, $\mathcal{K}^x$ is the complete conditioned- x response status, $\mathbf{y}^{r, \text{opt}}$ and $\mathcal{K}^{r, \text{opt}}$ are the requested conditioned- r companion and its coordinate-diagonal response or typed local/global unavailability, and $\mathbf{r}^{A, \text{parent}}$ is the immutable paired raw parent, $\mathcal{S}$ support, $\mathcal{I}$ influence, $\mathcal{F}$ typed flags/causes, $\mathcal{V}$ separated validity and eligibility inputs, $\mathcal{U}$ uncertainty/covariance availability, $\mathcal{P}$ one-way state/provenance, $\mathcal{D}$ diagnostics, and $\mathcal{C}$ atomic completion or required-output failure state.

3.5 Scientific design and learn–resolve–apply identities

For a locally Gaussian effective beam with angular covariance Σ and angular scan-velocity vector $\mathbf{v}$, the projected source-crossing time scale is

$$\sigma_t = (\mathbf{v}^T \Sigma^{-1} \mathbf{v})^{-1/2}. \quad (\text{SCI-RTC-EQ-023})$$

Equation (SCI-RTC-EQ-023) is a local Gaussian diagnostic. Scientific admission shall use the exact source-times-realized-response model over the declared beam, scan, source-class, and cadence domain.

A generic realized notch may be represented, where applicable, by

$$H_{\text{notch}}(z) = g \frac{\prod_{a=1}^{N_z} (1 - z_a z^{-1})}{\prod_{b=1}^{N_p} (1 - p_b z^{-1})}, \quad |p_b| < 1 \text{ for a causal-stability claim.} \quad (\text{SCI-RTC-EQ-024})$$

This form imposes no universal biquad. Exact realized coefficients, sample rate, frequency convention, phase/direction, state, edges, and precision are part of the response.

For an admitted FIR design family H_N , a selected order shall follow a declared constrained problem of the form

$$N_{\star} = \min_N \left\{ N : \begin{array}{l} \sup_{f \in \mathcal{P}} |H_N(f) - H_{\text{target}}(f)| \leq \epsilon_p, \\ \sup_{f \in \mathcal{S}} |H_N(f)| \leq \epsilon_s, \\ \text{phase/delay, alias, edge/guard, precision, and realizability predicates pass} \end{array} \right\}. \quad (\text{SCI-RTC-EQ-025})$$

The sets, metrics, tolerances, equality conventions, and family are owner-resolved; an approximate order formula is not universal authority.

For a causal linear-phase FIR with N taps at rate f_{samp} , its nominal interior group delay is

$$\tau_g = \frac{N-1}{2f_{\text{samp}}}. \quad (\text{SCI-RTC-EQ-026})$$

Where a temporal response has effective delay τ during a locally uniform scan, the corresponding scan-direction angular displacement is

$$\Delta\theta = \mathbf{v} \tau. \quad (\text{SCI-RTC-EQ-027})$$

For every learned RTC operation, the lifecycle is

$$\Lambda = \text{Learn}(\mathcal{L}, \mathbf{u}^{(0)}; \Theta^{\text{eff}}), \quad \Pi = \text{Resolve}(\Lambda, \Theta^{\text{policy}}), \quad \mathbf{u}^{\text{eval}} = \mathcal{A}_{\Pi}(\mathbf{u}^{(0)}), \quad \frac{\partial \Pi}{\partial j} = 0 \text{ during apply.} \quad (\text{SCI-RTC-EQ-028})$$

The final equality denotes plan immutability over apply slots; it is not an assumption that the realized operator is stationary at masks, edges, state boundaries, or other declared exceptions.

For bounded iterative notch-plan refinement with original admitted pair $\mathbf{u}^{(0)} = ((\mathbf{x}^A)^{\top}, (\mathbf{r}^A)^{\top})^{\top}$, initial fixed or identity plan Π_0 , actual attempt count A , configured maximum A_{max} , attempt index $a \in \{1, \dots, A\}$, and accepted-plan index k_a available at the start of attempt a ,

$$\begin{aligned} \mathbf{u}^{\text{eval},(0)} &= \mathcal{A}_{\Pi_0}(\mathbf{u}^{(0)}), \quad k_1 = 0, \\ \Lambda_a &= \text{Learn}(\mathcal{L}_a, \mathbf{u}^{\text{eval},(k_a)}, \Pi_{k_a}), \\ (\tilde{\Pi}_a, \delta_a) &= \text{Resolve}(\Lambda_a, \Pi_{k_a}, \Theta^{\text{policy}}), \\ k_{a+1} &= k_a + \mathbb{I}[\delta_a = \text{accepted}], \\ \delta_a = \text{accepted} &\implies \begin{cases} \Pi_{k_{a+1}} = \tilde{\Pi}_a, \\ \mathbf{u}^{\text{eval},(k_{a+1})} = \mathcal{A}_{\Pi_{k_{a+1}}}(\mathbf{u}^{(0)}) \end{cases}, \\ \delta_a \neq \text{accepted} &\implies \Pi_{k_{a+1}} = \Pi_{k_a} \text{ and no new evaluation is required,} \\ \mathbf{u}^{\text{final}} &= \mathcal{A}_{\Pi_K}(\mathbf{u}^{(0)}), \quad K = k_{A+1} \leq A \leq A_{\text{max}} < \infty, \\ \frac{\partial \Pi_k}{\partial j} &= 0 \text{ during every apply of accepted plan } k. \end{aligned} \quad (\text{SCI-RTC-EQ-029})$$

Equation (SCI-RTC-EQ-029) separates the actual finite attempt count from its configured maximum and from the accepted-plan sequence. It does not create artificial no-op attempts after an earlier stop. It defines the initial evaluation for nonidentity Π_0 , and states the default original-pair replay rule. The application- context projection publishes required conditioned x and, only when requested, conditioned r or its exact availability state. A sequential cascade from a preceding evaluation product requires separate explicit operator authority and complete propagation of every intermediate parent, response, state, phase, edge, support, covariance, and uncertainty.

3.6 Paired coordinates, leakage, and level shifts

For every native acquired detector occurrence, the upstream mapping authority shall produce one ordered pair from measured in-phase and quadrature coordinates before the separate ALIGN relation of equation (SCI-RTC-EQ-004),

$$\begin{bmatrix} x_{dm}^{\text{acq}} \\ r_{dm}^{\text{acq}} \end{bmatrix} = \mathcal{T}_{d,\zeta}(I_{dm}^{\text{acq}}, Q_{dm}^{\text{acq}}), \quad \mathcal{J}_{dm}^{xr,\text{acq}} = (d, m, \text{stream}, \zeta). \quad (\text{SCI-RTC-EQ-030})$$

The pair identity is exact. Missing either member is malformed required input; validity may nevertheless differ between the two members. An affine matrix, offset, or Jacobian is only a local representation over a declared domain of the generally nonlinear mapping $\mathcal{T}_{d,\zeta}$.

An admitted local optical response model may be written

$$\begin{bmatrix} \Delta x_d \\ \Delta r_d \end{bmatrix} = \begin{bmatrix} \alpha_d^x \\ \alpha_d^r \end{bmatrix} s_d + \begin{bmatrix} \eta_d^x \\ \eta_d^r \end{bmatrix}, \quad \alpha_d^x \neq 0. \quad (\text{SCI-RTC-EQ-031})$$

This is diagnostic authority, not permission to correct one member with the other or to assume $\alpha_d^r = 0$.

For a declared source class c and support-qualified estimator,

$$\begin{aligned} \epsilon_d^{(c)} &= \frac{\alpha_d^{r,(c)}}{\alpha_d^{x,(c)}}, & \Delta r_d &= \epsilon_d^{(c)} \Delta x_d + \eta_d^{(c)}, \\ \psi_d^{(c)} &= \text{atan2}\left(\alpha_d^{r,(c)}/s_{d,\zeta}^r, \alpha_d^{x,(c)}/s_{d,\zeta}^x\right) & \text{only under a declared commensurate metric.} \end{aligned} \quad (\text{SCI-RTC-EQ-032})$$

Atmosphere-derived and bright-source-derived estimates are distinct products with distinct parentage, support, validity, uncertainty, and scalar-versus-frequency-dependent status. The ratio carries the coordinate-unit ratio when x and r are not dimensionless under a common normalization. Comparison across mapping revisions is unavailable unless their units, scales, normalization, metric, and domains are compatible or an explicit transform has been applied.

For a declared level-shift event e , detector d has finite transition support $\mathcal{T}_{de}$ on the admitted aligned timing vector t_j^A . The v0.1 stable-plateau model is additive:

$$\begin{aligned} \mathcal{T}_{de} &= [t_{de}^-, t_{de}^+], & \Delta t_{de} &= t_{de}^+ - t_{de}^- > 0, \\ \tau_{de} &\in \mathcal{T}_{de}, & \tau_{de} &= \tau_e + \delta\tau_{de}, & |\delta\tau_{de}| &\leq \Delta\tau_e^{\text{coh}}, \\ \mathcal{I}_{de} &= \{j : t_j^A \in \mathcal{T}_{de}\}, \\ x_{dj}^A &= s_{dj}^x + b_{de}^- \quad (t_j^A < t_{de}^-), & x_{dj}^A &= s_{dj}^x + b_{de}^+ \quad (t_j^A > t_{de}^+), & \Delta b_{de} &= b_{de}^+ - b_{de}^-. \end{aligned} \quad (\text{SCI-RTC-EQ-033})$$

No plateau equation applies on $\mathcal{I}_{de}$. The network event identity, detector interval localization or conservative bound, physical-time width, timing-vector-derived sample support, coherence tolerance, separately measured x/r evidence or unavailability, and resulting segment boundaries belong to the realized record. No multiplicative gain or responsivity term is admitted.

For sufficiently supported stable pre/post plateau sets $\mathcal{G}_{de}^-, \mathcal{G}_{de}^+$, an admitted additive correction and reset condition may be represented by

$$\begin{aligned} \widehat{b}_{de}^\pm &= \text{Plateau}(x_{d,\mathcal{G}_{de}^\pm}^A; \Theta_P), & \widehat{\Delta b}_{de} &= \widehat{b}_{de}^+ - \widehat{b}_{de}^-, & \mathcal{G}_{de}^\pm \cap \mathcal{I}_{de} &= \emptyset, \\ x_{dj}^{A,\text{corr}} &= x_{dj}^A - \widehat{\Delta b}_{de} \mathbb{I}[t_j^A > t_{de}^+] \quad (j \notin \mathcal{I}_{de}), & \mathbf{s}_{j_g^+} &= R_g(\mathbf{s}_{j_g^-}; \Theta_R), \end{aligned} \quad (\text{SCI-RTC-EQ-034})$$

This displayed orientation translates the post-event plateau to the pre-event reference; a selected plan may use the opposite reference and sign. The transition cells remain flagged and unmodeled, not values made scientifically valid by the offset. If either required stable support or its selected quality/uncertainty criterion fails, $\widehat{\Delta b}_{de}$ and the correction are unavailable. R_g is ordinarily reset; carry across an accepted shift remains separately authorized continuity.

The actual bounded-refinement counts satisfy

$$0 \leq K \leq A \leq A_{\max} < \infty, \quad A = \#\{a : \text{Learn}_a \text{ was actually attempted}\}. \quad (\text{SCI-RTC-EQ-035})$$

On the common pair grid $\mathcal{G}_{\text{pair}}$, define the admitted raw pair and any separately available coordinate-specific affine baseline state by

$$\begin{aligned} \mathbf{u}_d &= \begin{bmatrix} \mathbf{x}_d^A \\ \mathbf{r}_d^A \end{bmatrix}, \\ \mathbf{b}_{d,k} &= \begin{bmatrix} b_{d,k}^x \\ b_{d,k}^r \end{bmatrix}, \\ \mathbf{u}_d^* &= \mathbf{u}_d - \mathbf{b}_{d,k} \quad \text{on stable segment } I_k. \end{aligned} \quad (\text{SCI-RTC-EQ-036})$$

The two affine components have separate estimators, availability, uncertainty, and action. This identity assumes neither equality nor permission to copy one component into the other.

Learn retains coordinate-specific and genuinely joint evidence as an ordered cause-preserving collection,

$$\mathcal{E}_{xr} = (E_x, E_r, E_{xr}), \quad E_c = (\mathcal{J}, c, \widehat{\theta}_c, \mathcal{S}_c, P_c, \zeta_c, U_c, Q_c, K_c, D_c, F_c), \quad c \in \{x, r, xr\}. \quad (\text{SCI-RTC-EQ-037})$$

Here $\mathcal{J}$ is exact occurrence identity, c is evidence origin, $\widehat{\theta}$ identifies the estimator output, $\mathcal{S}$ its support, P protection/Tune/mapping state, U uncertainty, Q quality, K causal class and direct/inferred state, D source/frequency

domain, and F its accepted, rejected, candidate, or unavailable disposition. This tuple is a minimum semantic inventory, not a storage schema.

For owner-designated hard classes after protection and admission predicates have been resolved, the pair action support is the cause-preserving union

$$B_{\text{pair}} = B_x^{\text{hard}} \cup B_r^{\text{hard}} \cup B_{xr}^{\text{hard}}, \quad \text{origin}(E_c) = c \text{ is retained.} \quad (\text{SCI-RTC-EQ-038})$$

This unions action support, not evidence values, causes, correction amplitudes, or generic flags. A diagnostic or spectral candidate is absent from B_{pair} until its stronger class-specific predicate passes.

Resolve composes the full evidence collection and policy into one immutable pair plan, and Apply consumes exactly that plan on both coordinates:

$$\Pi_{xr} = \text{Resolve}_{xr}(\mathcal{E}_{xr}, \Theta^{\text{policy}}), \quad (\mathbf{y}_d, \mathcal{R}_d) = \text{Apply}_{\Pi_{xr}}(\mathbf{u}_d), \quad \frac{\partial \Pi_{xr}}{\partial j} = 0 \text{ during Apply.} \quad (\text{SCI-RTC-EQ-039})$$

$\mathcal{R}_d$ is the realized cause/action/availability record. Evidence discovered after resolve belongs to a successor learn–resolve–apply attempt, never an in-place plan mutation.

On $\mathcal{A}_x \cap \mathcal{A}_r$, the canonical ordinary numerical action and its fixed-state Jacobian are

$$\mathbf{y}_d = (I_2 \otimes L_{\Pi}) \mathbf{u}_d^*, \quad J_{\text{pair}, \Pi} = I_2 \otimes L_{\Pi} = \begin{bmatrix} L_{\Pi} & 0 \\ 0 & L_{\Pi} \end{bmatrix}, \quad \frac{\partial \mathbf{y}^x}{\partial \mathbf{r}} = \frac{\partial \mathbf{y}^r}{\partial \mathbf{x}} = 0. \quad (\text{SCI-RTC-EQ-040})$$

L_{Π} includes identical masks, pair support, segments, transitions, guards, coefficients, state and reset, filter definition, order and precision, phase/delay, boundaries, downsampling, selected times, representative occurrences, cardinality, and grid. The x -only donor-recovery exception is not inserted into the r block: over its complete causal influence the conditioned- r value and response are typed unavailable on the unchanged grid. A distinct r operator would be a separately named descendant.

For an admitted joint input covariance on common available support, the fixed-state conditional propagation is

$$C_{y|\Pi} = (I_2 \otimes L_{\Pi}) C_{\text{in}} (I_2 \otimes L_{\Pi})^{\top}, \quad C'_{xr} = L_{\Pi} C_{xr} L_{\Pi}^{\top}. \quad (\text{SCI-RTC-EQ-041})$$

Available cross blocks are retained. Conditional fixed-state propagation, coordinate-specific affine-correction uncertainty, learned-parameter and selection uncertainty, donor-induced r unavailability, and total derivative availability remain distinct. The zero fixed-state cross response does not erase the valid data-dependent path $r \rightarrow E_r/E_{xr} \rightarrow \Pi_{xr} \rightarrow y_x$.

The support and availability states obey

$$\mathcal{A}_{\text{pair}} \subseteq \mathcal{G}_{\text{pair}}, \quad \mathcal{A}_x, \mathcal{A}_r \subseteq \mathcal{G}_{\text{pair}}, \quad \mathcal{A}_C \subseteq \mathcal{A}_x \cap \mathcal{A}_r, \quad \mathcal{A}_x \neq \mathcal{A}_r \text{ is permitted.} \quad (\text{SCI-RTC-EQ-042})$$

The first set records common pair action and ordinary-operator support, not joint numerical completeness. A required but unavailable affine correction removes the affected coordinate from $\mathcal{A}_x$ or $\mathcal{A}_r$ over the full downstream realized influence while preserving $\mathcal{G}_{\text{pair}}$ and the other coordinate’s independent availability.

4 Bounded assumptions and non-goals

SCI-RTC-ASM-001 – Capability.

V0.1 admits the exact paired raw readout coordinates $\mathbf{x}\mathbf{s}$ and $\mathbf{r}\mathbf{s}$. The x member is the ordinary science-conditioning path; the r member is a retained diagnostic coordinate and, when requested, an optional conditioned companion on exactly the conditioned- x grid. It is not an independently planned pipeline. Raw x and conditioned r are not themselves Stokes observables, and enabled polarimetry remains excluded until separately approved.

SCI-RTC-ASM-002 – Aligned-grid boundary.

The input grid, cadence, phase, mapping, gaps, synthesis/origin state, support, and scan partition are admitted ALIGN facts. Physical event timing, absolute timing accuracy, and a producer epoch may remain unavailable. RTC’s local numerical operator begins at this aligned pair; ALIGN remains upstream lineage and is composed only for a separately identified end-to-end response.

SCI-RTC-ASM-003 – Coordinate boundary.

Coordinate-dependent operations assume an admitted AST identity, frame, topology, detector binding, support, validity, and uncertainty where required. The absence of such authority does not license inference or clamping.

SCI-RTC-ASM-004 – Calibration boundary.

Calibration-factor production, target-atmosphere correction, beam meaning, uncertainty, and promotion belong to SCI-BEAM/SCI-CAL. RTC remains in raw $\Delta f/f$ through replacement, filtering, and sampling. Atmospheric templates are diagnostic evidence for leakage, Tune/readout health, shift and artifact assessment, and selection/review only; v0.1 does not subtract them from science x . SCI-CAL consumes only the exact RTC x product and applies absolute `flxscale` plus target atmosphere only after RTC.

SCI-RTC-ASM-005 – Selected-policy boundary.

Despike, level-shift detection, transition masks, segmentation, reset, plateau use, donor, mask, FIR, IIR, notch, atmospheric-template diagnostics, leakage, state, edge, non-finite, recovery, sampling, diagnostic, and failure choices are operation classes admitted across application contexts and governed by selected versioned policies. Admission does not mean execution; a class runs only when the resolved plan selects and numerically resolves it. Artifact evidence may be coordinate-specific, but detection and affected support are pair-coherent. Coordinate-valued corrections remain distinct; an x repair does not imply an r reconstruction. No current default or numerical threshold is promoted to a universal scientific constant by this contract.

SCI-RTC-ASM-006 – Conditional linearity.

Linear response and covariance identities condition on fixed paired-stream selection and realized state. At fixed state the paired conditioned response has identical ordinary diagonal blocks and both cross-coordinate numerical Jacobian branches are zero. Available joint covariance retains its cross-coordinate blocks even though numerical mixing is absent. Cross-coordinate covariance and shared selectors are retained in selection, learned-parameter, model, and cross-term uncertainty. Data-derived selectors make the unconditional operation generally nonlinear and nonstationary, and a total derivative may be unavailable at a selector boundary.

SCI-RTC-ASM-007 – Statistical input.

Moments or covariance are claimed only from a supplied statistical model with declared included and excluded components, correlations, and limitations. Absence does not imply diagonal, white, stationary, independent, or zero covariance, and a qualified partial claim need not pretend to be total.

SCI-RTC-ASM-008 – Learned-mode scope.

Every learned-then-fixed RTC operation uses one declared learning population, one resolved immutable plan, and one explicit compatibility domain. The first learned sampling scope produces one common observation plan. Per-array, per-scan, heterogeneous-cadence, noise-aware, and continuously adaptive sampling execution are successor scopes. Bounded iterative notch-plan refinement is admitted only as a finite outer sequence of actual attempts, bounded by a separately configured maximum, over complete immutable accepted plans; rejected attempts do not advance the accepted-plan sequence, unused maximum attempts are not artificial no-ops, and no parameter evolves within an apply pass.

SCI-RTC-ASM-009 – Analytical authority.

Exact analytical source-times-realized-filter transfer and phase-zero alias accounting define learned-plan admission. Deterministic vectors may falsify a realization, while source injections and observational studies are required for separately claimed performance or science-impact evidence.

SCI-RTC-ASM-010 – Compact representation.

Dense observation-sized response, covariance, or per-sample provenance is not required when a factorized, sparse, interval-based, or generative representation exactly reconstructs the operator, support, causes, and exceptions. Normal despike output may retain compact treatment state and population counts and characteristics without a complete per-event or per-donor manifest; detailed event fits and donor links may be a separately requested verbose or diagnostic product whose presence is numerically inert.

SCI-RTC-ASM-011 – Downstream authority.

RTC supplies causal inputs but does not own PTC cleaning/weights, VAL eligibility policy, MAP estimation, BEAM inference, empirical NOI products, MAP-003 response tracing, map filtering, source fitting, or fruit-loop recurrence. It reports the conditioned x/r pair and enough exact RTC facts for those consumers to apply their own policies; it does not select a PTC joint- r mode or a cross-package validity aggregation policy.

SCI-RTC-ASM-012 – Claim separation.

The contract is an algebraic and semantic authority only. It contains no claim that a present implementation conforms, that a representation is faithful, that observational performance is adequate, or that a mode is ready for production.

4.1 Explicit non-goals

This revision does not derive ALIGN timing, AST coordinates, SCI-BEAM or SCI-CAL calibration, PTC, VAL, MAP, or FLT estimators, source fitting, or recurrence; authorize absolute-timing or astrometric corrections; redesign mature despiking/FIR/IIR/notch numerics; choose universal numerical thresholds; authorize online adaptation; activate polarimetry; authorize atmospheric or common-mode template subtraction from science x ; authorize automatic r -to- x correction, r calibration, use of r as an x donor, or a donor-based or otherwise invented r repair; decide how PTC uses conditioned r or how SCI-VAL aggregates cross-package validity; or authorize a default joint- r mode; gain-, responsivity-, or general response-changing level-shift model; prescribe source classes, files, storage schemas, or product names; authorize per-array/per-scan cadence; inspect implementation; perform validation or reductions; or assert production status.

5 Sequential normative requirements

The following identifiers are stable within v0.1/r0.12. Conformance is requirement-by-requirement; satisfying a neighboring requirement does not waive another one.

Area	ID	Normative statement
Boundary	SCI-RTC-REQ-001	RTC shall admit only exact aligned xs/rs pairs whose signals are raw, uncalibrated detector coordinates and whose ordered pair identity, assigned grid, support, origin/synthesis state, and independently declared validity are present; a missing partner is malformed required input.
Boundary	SCI-RTC-REQ-002	Every requested RTC application context shall declare requested use, paired input and interval, x/r sign and reference/baseline, valid domain, named consumers, and constraints while retaining raw coordinate identity; a context label shall neither restrict admitted operation classes nor give either member a calibrated or Stokes meaning.
Boundary	SCI-RTC-REQ-003	Every RTC application context shall produce conditioned raw x as its required numerical output and retain the exact paired raw r parent plus every causal r diagnostic, selector, segmentation, and resolved-plan use in the consumer-neutral atomic bundle. When requested, conditioned r shall be an optional paired companion under the same context, plan, realized record, and selected grid; it is not a separately optimized pipeline.
Boundary	SCI-RTC-REQ-004	A calibrated detector signal shall be a distinct downstream SCI-CAL product in the top-of-atmosphere, point-source-equivalent mJy-per-fixed-nominal-beam convention, with admitted factor, atmosphere, detector, validity, uncertainty, and provenance state.
Boundary	SCI-RTC-REQ-005	RTC shall preserve the exact paired raw parent, conditioned- x plan/response/support, any requested conditioned- r availability/response/support, factor authority, atmospheric diagnostic lineage, and distinct SCI-CAL atmosphere lineage required downstream; atmospheric templates may inform diagnostics, selectors, or review but shall not be numerically subtracted from science x in v0.1.
Identity	SCI-RTC-REQ-006	Observation, Tune, array, network/interface, detector occurrence, APT binding, native row, aligned slot, scan, science interval, context interval, output slot, stage, and product identity shall remain distinct and immutable.
Identity	SCI-RTC-REQ-007	The ordered detector and sample shapes, index bases, native-to-aligned mapping, aligned-to-output map, scan membership, and memory order shall be explicit; indices shall not serve as cross-artifact detector identity.
Identity	SCI-RTC-REQ-008	RTC shall preserve the admitted ALIGN cadence, phase, rate, gaps, support, and synthesis/origin state without claiming physical integration-event timing or absolute epoch authority.

Area	ID	Normative statement
Identity	SCI-RTC-REQ-009	A coordinate-dependent operation shall consume explicit AST identity, frame, topology, detector binding, support, and validity required by its resolved plan, with no inference, clamping, or implicit frame conversion.
State	SCI-RTC-REQ-010	RTC application context, immutable resolved plan, and realized record shall be distinct one-way objects; requested, effective, observation-resolved, learned, resolved, and realized substates may refine them, but no later object or substate shall overwrite an earlier one.
State	SCI-RTC-REQ-011	Outer, inner, full, mini, diagnostic, simulated, processed, and any other produced stage roles shall have distinct immutable identities and explicit parent/processing links without requiring duplicate computation.
Policy	SCI-RTC-REQ-012	Despike, spike-evidence, level-shift learning, segmentation, plateau-assessment, leakage-diagnostic, donor, replacement, mask, filter, state, edge, non-finite, recovery, sampling, flag, diagnostic, and fallback classes are admitted across RTC application contexts. Evidence may be coordinate-specific, but pathology detection and affected-support decisions shall be pair-coherent. Availability shall not be confused with enabled, selected, numerically resolved, or scientifically qualified; every executed action shall be governed by an admitted versioned resolved plan rather than an undocumented default.
Order	SCI-RTC-REQ-013	RTC shall begin at the admitted upstream-aligned paired raw parent, perform spike-candidate masking/downweighting for level-shift learning on that original pair, resolve finite transition support and stable plateaus, apply any separately selected and valid coordinate-specific additive plateau corrections, then apply compatible x donor replacement within stable segments. Ordinary filtering and phase-zero sampling shall use the identical paired operator, while the x donor exception shall make affected r support typed unavailable without reconstruction. SCI-CAL consumes only x as in equation (SCI-RTC-EQ-004); no calibrated RTC branch or second ALIGN application is authorized.

Area	ID	Normative statement
Donor	SCI-RTC-REQ-014	A raw donor q to raw target d using the detector-static comparison coordinate $\chi_i = \phi_i x_i$ shall use ϕ_q/ϕ_d only when both flxscale factors are valid for the exact occurrences under the same compatible convention/domain and $\phi_d \neq 0$.
Donor	SCI-RTC-REQ-015	The raw donor route in REQ-014 shall use flxscale and shall neither require nor restore a scientific role for legacy responsivity .
Donor	SCI-RTC-REQ-016	When REQ-014 is not available, replacement shall fail under that route unless a separate donor-to-target authority declares direction, quantity/unit domain, support, uncertainty, validity, and provenance.
Donor	SCI-RTC-REQ-017	When despiking is selected, RTC shall actually apply the selected replacement or recovery to each accepted target x cell, or record the selected explicit failure/no-correction disposition. The pathology and affected support shall apply to the paired occurrence, while an x repair shall not require numerical r replacement. The resolved detector, target, eligibility, topology, score/tie, donor-weight/time, source-mask/flag, and fallback policies shall be reconstructible. Normal production output shall retain compact treatment state and useful spike-population counts and characteristics; complete event-by-event fits and realized donor links are required only for a separately selected verbose or diagnostic product.
Donor	SCI-RTC-REQ-018	A replaced target shall retain target identity, link every donor and factor, preserve the pair-level replacement cause, and expose correlations from reused or shared donors; finiteness shall not create independent exposure. No donor-derived r value shall be invented when an r -domain correction is unavailable.
Influence	SCI-RTC-REQ-019	An output whose exact representative source occurrence $\rho_{dn} = (d, Mn)$ is ALIGN-synthesized shall be universally excluded as an independent detector measurement; nonrepresentative synthesized influence shall preserve its numerical effect, typed cause, and support for the downstream consumer's declared eligibility policy.
Influence	SCI-RTC-REQ-020	An output whose exact representative source occurrence $\rho_{dn} = (d, Mn)$ is RTC-replaced shall be universally excluded as an independent detector measurement; nonrepresentative replacement influence shall preserve its numerical effect, typed replacement cause, declared donor-influence class, and support for the downstream consumer's declared eligibility policy.

Area	ID	Normative statement
Filter	SCI-RTC-REQ-021	Every enabled response-changing stage shall record chronological order, exact full-precision coefficients or exact digest, coefficient convention, normalization, numeric precision, rate, direction, state, context, coordinate applicability, mask, edge, non-finite policy, exceptions, and the coherent segment bounded by gaps, resets, or level shifts. When an ordinary shared stage applies to requested conditioned r , its parameters, sampling phase, and state boundaries shall match the paired conditioned- x plan.
Filter	SCI-RTC-REQ-022	An FIR claim shall expose exact impulse support, DC gain, phase convention, and normalization; a label such as “normalized” shall not substitute for the coefficient sum or response.
Filter	SCI-RTC-REQ-023	An IIR or notch claim shall expose an exact recurrence or state-space equivalent, realized poles/zeros where applicable, initial/final state, reset boundary including admitted level shifts, transient/guard rule, precision, and stability status; reset is ordinary at an accepted shift, and carry is permitted only as separately authorized continuity with explicit response and residual-contamination criteria. Paired conditioned coordinates shall share the resolved boundary/reset disposition wherever that stage applies.
Mask	SCI-RTC-REQ-024	A processing mask shall be represented as an operator control, including geometry or selection, boundary convention, dilation, denominator/segment behavior, state interaction, and pair-coherent affected support; it shall not be treated as acquisition validity or confidence.
Mask	SCI-RTC-REQ-025	Invalid, unavailable, wrong-frame, out-of-domain, or ambiguously bound coordinates shall not become outside-source samples and shall fail the affected required coordinate operation.
Finite	SCI-RTC-REQ-026	A non-finite required sample, factor, coefficient, coordinate, or filter state shall be rejected with a typed cause, marked unavailable with that cause over its exact transitive influence, or handled by an explicitly authorized prior replacement/recovery rule; it shall not be silently coerced to zero or encoded only as an undifferentiated generic non-finite. Optional conditioned- r unavailability shall remain local where possible and shall not invalidate otherwise valid conditioned x .
Edge	SCI-RTC-REQ-027	Boundary extension, incomplete context, filter guards, IIR recovery, short/empty scans, and output selection shall be explicit; an edge or incomplete-state cell shall not be labeled an ordinary interior sample.

Area	ID	Normative statement
Sampling	SCI-RTC-REQ-028	V0.1 downsampling shall use only phase-zero point selection from the final pre-decimation streams. When conditioned r is requested, $y^{x,\text{RTC}}[n] = v^{x,\text{preD}}[Mn]$ and available $y^{r,\text{RTC}}[n] = v^{r,\text{preD}}[Mn]$ shall occupy exactly the same output grid, cardinality, selected times, and representative occurrence (d, Mn) on the raw aligned paired parent. Arithmetic-mean or other block-aggregate downsampling is excluded.
Sampling	SCI-RTC-REQ-029	Each coherent segment shall record M , zero phase, input/output rates and grids, the cardinality of equation (SCI-RTC-EQ-011), selected-point representative time, full transitive support, edge disposition, and flag/influence aggregation.
Sampling	SCI-RTC-REQ-030	Any alias-free or bounded-alias claim shall include the exact realized prefilter and all folded bands of equation (SCI-RTC-EQ-014), with a declared input band, metric, bound, and unavailable state; point selection alone shall not be called anti-alias filtering.
Sampling	SCI-RTC-REQ-031	Fixed mode shall resolve an admitted factor/filter/rate/compatibility plan before application and record it independently of realized output.
Learned	SCI-RTC-REQ-032	Every learned RTC operation shall preserve distinct requested, effective, observation-resolved, learned-evidence, resolved-plan, and applied states and shall not advertise a plan before admitted evidence and numeric policy are complete.
Learned	SCI-RTC-REQ-033	Learned resolution shall choose the largest admitted integer factor only after astronomical-transfer, exact phase-zero alias, sampling, filter-realizability, and downstream-compatibility constraints all pass using the smallest admitted beam and maximum valid in-scan speed.
Learned	SCI-RTC-REQ-034	The first learned applied scope shall use one common immutable observation cadence and filter across arrays and scans; percentile speed shall remain diagnostic rather than safety authority.
Learned	SCI-RTC-REQ-035	Apply shall consume exactly one immutable resolved plan and shall not retune, adapt, or silently fall back; an unavailable or rejected plan shall follow the selected failure/fallback policy before apply begins.
Learned	SCI-RTC-REQ-036	Restart shall restore a state-complete resolved plan, verify exact plan/input compatibility, and reject stale or incompatible beam, speed, scan, coefficient, rate, phase, application context, consumer binding, or parent identity.
Response	SCI-RTC-REQ-037	The atomic bundle shall contain the complete RTC-local realized conditioned- x response and, when conditioned r is requested, the complete coordinate-diagonal conditioned- r response on its available support, or a typed unavailable state for either. Optional end-to-end response to native acquisition coordinates shall be separately identified and explicitly composed with admitted upstream ALIGN response. A partial filter kernel or silently repeated ALIGN operator shall not be labeled the conditioned response.
Response	SCI-RTC-REQ-038	Selection, detector mixing, masking, finite boundaries, state, or multirate behavior shall use a local/factorized detector-time response or truthful unavailability rather than an unjustified scalar transfer. Fixed-state $x \leftarrow r$ and $r \leftarrow x$ numerical responses are both zero; cross-coordinate selection dependence is separate, and downstream SCI-CAL response remains a separate composition.
Response	SCI-RTC-REQ-039	A scalar LTI response shall be claimed only on a stated uniform, fixed-state, detector-separable, translation-invariant interior domain with no data-derived selector or undeclared rate change.
Response	SCI-RTC-REQ-040	Where defined, response shall distinguish amplitude, power, phase, group delay, sampling rate, Fourier convention, alias images, support, and response uncertainty; undefined phase/delay at a response zero shall remain unavailable.
Response	SCI-RTC-REQ-041	RTC-local output support shall be expanded transitively from the admitted aligned grid through replacement, every filter, state boundary, edge rule, and phase-zero selection; optional end-to-end support shall additionally compose admitted upstream ALIGN support. A selected center index shall not stand in for full support.

Area	ID	Normative statement
Statistics	SCI-RTC-REQ-042	At fixed realized state, conditioned- x mean and covariance shall follow equations (SCI-RTC-EQ-016a)–(SCI-RTC-EQ-016b), so only Σ_{xx} enters that coordinate’s direct propagation. When conditioned r and an admitted joint covariance are available, the pair shall follow equation (SCI-RTC-EQ-036) and preserve the available xr/rx blocks. Every claim shall name its conditioning and included/excluded cross-coordinate, donor-reuse, shared-selector, overlap, learned-parameter, model, diagnostic, and cross-term effects.
Statistics	SCI-RTC-REQ-043	When selection depends on modeled random data, each uncertainty claim shall state whether selection bias, selection covariance, and correlations with retained terms are included, excluded, or unavailable; they shall not be set to zero merely by conditioning on the realized selector.
Statistics	SCI-RTC-REQ-044	Every RTC covariance or uncertainty claim shall explicitly name the components and correlations it includes and excludes, their support and basis, and the availability of calibration, atmosphere, beam, donor-scale, coefficient, cadence/phase, response, selection, and model nuisance terms and applicable cross terms.
Statistics	SCI-RTC-REQ-045	Unknown covariance components or correlations shall remain unavailable and shall not be replaced by diagonal, white, stationary, independent, zero, or scalar-weight assumptions. Truthfully labeled conditional or component-limited claims are permitted; only declared complete coverage may be called full precision or total uncertainty.
Validity	SCI-RTC-REQ-046	Acquisition support, coordinate-specific evidence, independent numerical validity of conditioned x and r , pair-level pathology/affected support, pair identity, ALIGN synthesis, numerical computability, coordinate validity, flags/causes, operator masks, replacement, transitive influence, complete support, response status, consumer eligibility, uncertainty availability, and provenance shall remain separate states.
Validity	SCI-RTC-REQ-047	A flag shall describe a cause and shall not automatically become a mask, invalidity, weight, or universal eligibility decision; aggregation shall preserve every contributing cause required by the selected policy.

Area	ID	Normative statement
Output	SCI-RTC-REQ-048	RTC shall produce the consumer-neutral atomic bundle of equation (SCI-RTC-EQ-022): conditioned x , requested conditioned r or its typed availability state, the immutable paired raw r parent, and every required common diagnostic/selector/lineage field. Consumers bind the members and optional diagnostic detail they need; neither conditioned x TOD alone nor an independently emitted r diagnostic is the complete required product.
Output	SCI-RTC-REQ-049	Required malformed identity/state or missing authority shall propagate to atomic required-output failure; optional unavailable provenance detail shall be typed and numerically inert.
Output	SCI-RTC-REQ-050	Compact provenance shall reconstruct the exact native IQ-to- x/r mapping, upstream ALIGN relation, admitted aligned pair identity, RTC-local input/parent identity, detector binding, grids and intervals, conditioned- r request and local/global availability, downstream CAL handoff authority, resolved donor policy and compact spike treatment/population summaries, level shifts/segments/resets, ordered coordinate applicability, filters, masks, coefficients/state/edges, sampling, local and optional end-to-end response/uncertainty status, exceptions, counts, and the one-way application-context/plan/record lifecycle. Detailed spike-event and donor-link manifests are separately requested diagnostic detail.
Output	SCI-RTC-REQ-051	Each diagnostic, including spike-population, raw and conditioned leakage, and level-shift products, shall name estimator, coordinate stage, mapping revision, unit, response, support, mask/selection, validity, availability, scalar-versus-frequency status where applicable, and whether it is inert, advisory, or enters the selected policy; toggling optional event-by-event or donor-level diagnostic detail shall not change values, flags, influence, eligibility, cardinality, response, or failure.
Consumer	SCI-RTC-REQ-052	Beyond the direct exclusions in REQ-019–020, RTC shall supply cause-preserving transitive influence rather than a universal downstream eligibility bit; each consumer shall bind the exact RTC parent, declare its influence-to-eligibility policy, and shall not erase causes or strengthen unavailable timing, coordinate, calibration, response, covariance, or eligibility claims.
Consumer	SCI-RTC-REQ-053	PTC-disabled terminal RTC output shall be a valid requested state when selected; no PTC or map product shall be implied, while any requested downstream product shall require the exact admitted RTC bundle.
Claims	SCI-RTC-REQ-054	Algebraic contract correctness, implementation conformity, representation fidelity, observational performance, science-impact qualification, and production readiness shall be stated and assessed as separate claims.

Area	ID	Normative statement
Lifecycle	SCI-RTC-REQ-055	Every learned operation shall name its learning population, estimator, support, selection, exclusions, quality, uncertainty, validity, and compatibility domain; same-observation learning shall be identified as data-dependent.
Lifecycle	SCI-RTC-REQ-056	Resolve shall convert evidence and fixed policy into one immutable plan containing every applicable sequence, family, coefficient identity, rate/convention, order/taps, band/notch parameter, applicability, mask, donor, edge/guard/reset, decimation/timing, response/support, uncertainty, validity, and downstream-compatibility field.
Lifecycle	SCI-RTC-REQ-057	A plan shall be resolved only when every required predicate and tolerance passes; missing evidence, missing tolerance, rejected evidence, or unavailable compatibility shall not be treated as pass.
Lifecycle	SCI-RTC-REQ-058	Any coefficient or operator parameter that evolves during apply shall be governed by a separately authorized online-adaptive estimator contract; it shall not be represented as learned-then-fixed filtering.
Design	SCI-RTC-REQ-059	Every despoke or spike-evidence, level-shift-learning, segmentation, plateau-assessment, leakage-diagnostic, donor, mask, notch, low/high/band-pass, FIR, IIR, decimation, edge/state, or non-finite operation shall declare scientific purpose, signal/contaminant model, mathematical operator, design parameters, learn/resolve/apply behavior, astronomical and calibration consequences, validation/falsification, and unavailable states.
Design	SCI-RTC-REQ-060	Filter admission shall use the exact source-times-realized-response over the declared scan-speed/direction, projected-beam, source-class, prior-response, and cadence domain; $f \sim v/\theta$, a scalar FWHM, or a nominal cutoff shall remain pedagogical only.
Notch	SCI-RTC-REQ-061	Every notch shall resolve center, rate/frequency convention, width or Q, depth, family/order, poles/zeros where applicable, phase/group delay, state, transient/guard, applicability, learned uncertainty, drift tolerance, and combined response for overlaps.
Bands	SCI-RTC-REQ-062	Every admitted low-, high-, or band-pass shall resolve pass/transition/stop bands, ripple, attenuation, phase/delay, family, order/taps, precision, edges/guards, retained astronomical modes, and unavailable modes; no class label shall substitute for exact response.

Area	ID	Normative statement
FIR	SCI-RTC-REQ-063	FIR order or tap count shall follow owner-resolved transfer, alias, phase, support, guard, precision, and realizability constraints such as equation (SCI-RTC-EQ-025); an arbitrary integer or approximate family formula shall not be scientific authority.
Donor	SCI-RTC-REQ-064	Donor replacement shall be identified as a continuity surrogate at the target identity and shall separately establish numerical convention transfer, donor admissibility, and scientific eligibility; compatible <code>flxscale</code> shall not be evidence of common sky signal.
Donor	SCI-RTC-REQ-065	Donor scaling in a Beammap application context shall use an explicitly identified prior APT or other pre-existing calibration authority and shall not use a new <code>flxscale</code> derived from the same Beammap conditioning lineage.
Sampling	SCI-RTC-REQ-066	Decimation factor and prefilter shall be justified from cadence, exact response, folded-band alias bounds, smallest projected crossing time, maximum valid scan speed, source class, timing/coordinate accuracy, and downstream compatibility.
Timing	SCI-RTC-REQ-067	Each plan shall resolve temporal registration through zero phase, shifted output time, coordinate correction, response-aware downstream use, or an approved bound on residual centroid/PSF effect; selected grid time alone shall not imply response centering.
CAL	SCI-RTC-REQ-068	Calibration transfer between Beammap and target processing shall bind the complete RTC plans and require matching response or explicit calculation, renormalization, or observational evidence; identical operation labels or preserved units are insufficient.
Validation	SCI-RTC-REQ-069	Validation of a notch, broad-band/FIR, donor, or decimation plan shall vary its scientific design parameters over the admitted scan/beam/source domain and measure astronomical transfer, phase/centroid, PSF/morphology, ringing, support/edge loss, covariance, and alias or interference residual as applicable.
Claims	SCI-RTC-REQ-070	No plan shall be called science-qualified solely because coefficient algebra, optimization, serialization, or deterministic operator fixtures pass.

Area	ID	Normative statement
Iteration	SCI-RTC-REQ-071	Iterative notch-plan refinement shall use a finite owner-approved maximum attempt bound and stopping policy; actual attempt count, configured maximum, and accepted-plan count shall remain distinct, unused maximum attempts shall not become artificial no-ops, and an unbounded repeat-until-satisfactory process is unavailable.
Iteration	SCI-RTC-REQ-072	Every apply of accepted plan Π_k shall consume exactly that immutable plan and shall not change coefficients, notch parameters, policy, masks, donor rules, decimation, or state conventions.
Plan	SCI-RTC-REQ-073	Every proposed successor $\tilde{\Pi}_a$ shall be a complete cumulative plan that dispositions all prior and proposed notches and records all stages, order, coefficients/precision, state, masks, edges/guards, intervals, decimation/timing, response/support, uncertainty, validity, and compatibility; rejection shall not assign it an accepted-plan index.
Replay	SCI-RTC-REQ-074	The default evaluation and final production rule shall apply each complete accepted plan to the original admitted pair $\mathbf{u}^{(0)}$ as in equation (SCI-RTC-EQ-029); exploratory attempts shall not accumulate hidden rounding, padding, state, duplicate-filter, mask, flag, or parent effects. Product projection shall publish required conditioned x and requested conditioned r or its exact availability state without removing raw- r evidence from learning.
Cascade	SCI-RTC-REQ-075	A cascade from a preceding filtered product shall require explicit scientific-operator authority and immutable intermediate parents, with total response, state, phase, edges, support, covariance, uncertainty, and sequential parentage retained.
Evaluate	SCI-RTC-REQ-076	Each successor attempt shall learn from the current accepted-plan evaluation and assess intended line attenuation and residual, parameter error/drift, predicted-versus-measured response, stability, ringing/transient, edge/guard loss, phase/delay, source amplitude/centroid/PSF, extended transfer, covariance, and support where applicable.

Area	ID	Normative statement
Candidate	SCI-RTC-REQ-077	A successor line shall be admitted only after comparison of original, predicted, and conditioned spectra, cumulative response, line uncertainty, astronomical spectrum, transfer/alias budgets, and detector/network/array/scan/interval consistency excludes notch-edge, ringing, transient, leakage, normalization, numerical, and response-floor artifacts.
Budget	SCI-RTC-REQ-078	Successor admission shall preserve owner-approved cumulative interference improvement, removed bandwidth, astronomical transfer, source amplitude/centroid/PSF, phase/registration, ringing/transient, guard/support, stability, cadence/response compatibility, uncertainty, and provenance budgets.
Stability	SCI-RTC-REQ-079	Plan stability shall compare the complete line set and matching identity, centers, widths/Q, depths, family/order, precision, applicability, state/reset, masks/guards, broad-band stages, decimation/timing, response/support, and compatibility over the required consecutive accepted-plan comparisons.
Stop	SCI-RTC-REQ-080	Every attempt shall record one typed accepted, rejected, or stop disposition plus its cause; maximum-attempt, oscillating-plan, and nonconvergent-parameter terminations shall not be labeled convergence or plan stability, and rejection shall leave the last accepted plan available unless policy says otherwise.
Lineage	SCI-RTC-REQ-081	Every attempt shall preserve attempt index, current accepted-plan index, original admitted pair, previous accepted plan, learning parent/population and PSD definition, original/predicted/measured paired spectra, proposal and disposition, predicates/tolerances, any newly accepted complete plan/evaluation product, cumulative response/astronomical diagnostics, stop cause, product projection, and final plan identity in one-way state.
Restart	SCI-RTC-REQ-082	Serialization and restart shall reproduce the final selected complete plan and original-pair replay rule exactly and reject incompatible original pair, rate, learning population, parent, coefficient, interval, state, or policy identity.

Area	ID	Normative statement
Pair	SCI-RTC-REQ-083	Every required admitted aligned occurrence shall contain both x_{dj}^A and r_{dj}^A with exact common detector, sample, stream, IQ-mapping revision, and ALIGN lineage; a missing member shall fail required-input admission rather than be synthesized, copied, or zero-filled.
Pair	SCI-RTC-REQ-084	Paired identity shall remain exact through selection, segmentation, coordinate-diagonal conditioned- x/r filtering/sampling where applicable, bundle construction, serialization, and restart. Retaining raw r parentage or sharing a grid does not apply x -specific repair to r , and reordering or independently joining members is not equivalent.
Mapping	SCI-RTC-REQ-085	The upstream native $IQ \rightarrow xr$ mapping and subsequent ALIGN relation shall expose distinct authority, revision, transform identity, any local coefficient/offset/Jacobian convention, applicability, validity, support, and provenance for each admitted occurrence; RTC shall consume the resulting aligned pair rather than infer, linearize globally, repair, or reapply either upstream mapping.
Validity	SCI-RTC-REQ-086	Numerical validity of conditioned x and r shall be represented independently while retaining pair-coherent pathology causes and affected support. Invalid or unavailable r , locally or globally, shall not invalidate, replace, alter, or validate otherwise valid conditioned x ; scientific eligibility remains downstream policy.
Leakage	SCI-RTC-REQ-087	Optical response in r shall not be assumed zero. Any leakage claim shall identify source class, estimator, support, selection, validity, uncertainty, x/r units or scales, mapping revision, normalization, metric, Tune/loading/epoch domain, ratio dimensionality, and response convention.
Leakage	SCI-RTC-REQ-088	An atmosphere-derived x/r leakage diagnostic shall identify the atmospheric template or population, temporal/frequency support, masks, estimator, quality, uncertainty, applicability, shared-data dependence, noise-in-both-coordinates treatment, and detector self-inclusion rule independently of any bright-source result; correlation or direct noisy r -on- x regression alone is not an unbiased coefficient claim.
Leakage	SCI-RTC-REQ-089	A bright-source-derived x/r leakage diagnostic shall identify the source, model, spatial/temporal support, masks, estimator, quality, uncertainty, and applicability independently of any atmospheric result.
Leakage	SCI-RTC-REQ-090	Atmosphere-derived and bright-source-derived leakage products shall retain separate parentage and shall not be averaged, substituted, or promoted as mutually validating without an explicitly authorized comparison model.

Area	ID	Normative statement
Leakage	SCI-RTC-REQ-091	Each leakage product shall state whether it is scalar, band-limited, or frequency dependent and shall provide the corresponding response/support and coordinate-comparison compatibility, or a typed unavailable state. A ratio or angle shall not be compared across detectors, Tunes, loading states, epochs, or mapping revisions unless coordinate conventions are compatible or an explicit transformation and metric are applied.
Boundary	SCI-RTC-REQ-092	V0.1 shall not automatically correct x using r , correct r using x , calibrate r , use either coordinate as a donor for the other, or treat one valid coordinate as a substitute for the other invalid coordinate.
Statistics	SCI-RTC-REQ-093	Any joint x/r selector, estimator, learned parameter, or diagnostic shall preserve cross-coordinate covariance, exact common index grid, pair-action support, coordinate-local availability, joint covariance support, shared selection, and unavailable components as distinct states; independence shall require evidence. Such dependence shall not be misreported as a fixed-state $r \rightarrow x$ numerical branch, and a total derivative may be unavailable at selector boundaries.
Shift	SCI-RTC-REQ-094	Every accepted level-shift disposition shall retain compact event identity and treatment state, network timing reference, detector-specific finite transition interval or conservative bound in physical time, timing-vector-derived pair-level affected support, coherence tolerance, coordinate-specific x and r evidence or unavailability, coordinate-specific additive-offset status, quality/uncertainty status, detection state, and cause. Normal production output may summarize event populations; event-by-event fits and detailed diagnostics are required only when the selected verbose/diagnostic product requests them.
Shift	SCI-RTC-REQ-095	Level-shift learning shall use the original aligned paired parent while identified isolated-spike candidates are excluded, masked, or robustly downweighted without donor substitution. Shift boundaries and pair-level affected support shall be resolved before any donor replacement; later x replacement shall remain within a resolved stable segment, no donor relation shall cross an unresolved or accepted shift boundary, and no r donor repair is implied.
Shift	SCI-RTC-REQ-096	A selected level shift shall determine or conservatively bound a finite transition interval in physical time and derive its pair-level affected sample support from the applicable timing vector. Those cells shall be explicitly flagged as level-shift-transition affected in both coordinates, excluded from applicable plateau estimators, assigned to neither plateau, and remain scientifically unmodeled after any coordinate-specific offset correction. Any downstream propagated influence or guard shall remain distinguishable from the underlying physical transition support.
Shift	SCI-RTC-REQ-097	A selected level shift shall delimit coherent paired processing segments and reset applicable filter or estimator state as the ordinary behavior. Carry across the boundary is permitted only as separately authorized continuity with an explicit response and residual-contamination criterion; donor links shall never cross it.
Plateau	SCI-RTC-REQ-098	Any plateau diagnostic or additive-offset estimate shall declare coordinate, segment, estimator, stable support on each side, transition exclusion, masks, uncertainty, quality, validity, reference plateau, and whether it is inert, advisory, or a selected policy input.

Area	ID	Normative statement
Shift	SCI-RTC-REQ-099	Multiple level shifts shall retain ordered distinct event identities, including overlapping guards and short intervening segments, and shall follow a selected conflict and minimum-segment policy.
Leakage	SCI-RTC-REQ-100	Raw-pair and conditioned-pair optical-leakage diagnostics shall be separately identified. When coordinate-specific preprocessing or filtering changes a leakage estimate, each product shall declare its stage, mapping revision, units, common comparison domain, support, response, validity, uncertainty, and source class; a resolved linear filter shall propagate nonzero optical response rather than erase it.
Plateau	SCI-RTC-REQ-101	A resolved RTC plan may estimate one additive difference between sufficiently stable pre/post x plateaus and translate one valid plateau to the declared reference plateau. The plan shall specify estimator, required support, quality/uncertainty criteria, reference plateau, application direction, and failure behavior. It shall not apply the offset to transition support or make that support valid, and it shall not introduce a gain, responsivity, or other response-change parameter.
State	SCI-RTC-REQ-102	Pair admission, conditioned- r request/availability, mapping, spike-candidate evidence, level-shift learning, physical transition support, propagated influence, segmentation, coordinate-specific additive-offset estimation/application, post-segmentation x replacement, reset, plateau evaluation, source protection, and raw/conditioned leakage diagnostics shall expose requested, effective, resolved, realized, unavailable, and failed states with typed causes as applicable.
CAL	SCI-RTC-REQ-103	SCI-CAL shall receive only the exact conditioned x member plus the complete paired RTC bundle needed for traceability; raw or conditioned r shall remain in its declared detector-coordinate unit/sign/reference and shall not receive SCI-CAL calibration or target-atmosphere correction.
Atm.	SCI-RTC-REQ-104	Atmospheric templates in RTC v0.1 shall be used only for leakage estimation, Tune/readout-health diagnostics, level-shift or artifact evidence, and declared selection/review inputs; they shall not be subtracted from science x . Any future numerical common-mode or atmospheric-removal estimator requires separate PTC or successor authority.
Context	SCI-RTC-REQ-105	Every resolved application-context plan shall disposition paired mapping, conditioned- r request and availability, x/r validity, retained raw r parentage, leakage diagnostics, source protection, level shifts, segmentation/reset, and diagnostic atmosphere treatment. All v0.1 operation classes remain admitted across contexts until evidence or explicit policy disqualifies them; only operations explicitly selected and numerically resolved execute. Consumers bind the same consumer-neutral bundle, and subsequent CAL, PTC, VAL, MAP, and FLT use follows their own contracts.

Area	ID	Normative statement
Plateau	SCI-RTC-REQ-106	A plateau shall be retained for a requested application context only when it contains enough valid support for every required operation and estimator in that context; support shall be evaluated against masks, source/atmosphere content, filter context/state, PSD/notch learning, source geometry, cadence, and downstream needs as applicable, not sample count alone. If stable pre/post support is insufficient for a defensible additive offset, RTC shall not invent one; the event remains a segment boundary and the resolved plan shall retain or reject each usable plateau explicitly.
Leakage	SCI-RTC-REQ-107	Stable plateaus immediately before and after an accepted level shift shall have their optical-leakage or more general paired optical-response states compared on declared joint comparison support where both required coordinates, responses, and statistical components are available, or the comparison shall be explicitly unavailable when adequate optical excitation or joint support is absent; evidence of a material response change shall make additive correction unavailable and shall not trigger a fitted gain-change model in v0.1. Gain- or response-changing interpretation remains successor scope unless observational evidence reopens this assumption.
Output	SCI-RTC-REQ-108	Requested conditioned r shall use RTC's paired-companion extension under the same application context, resolved plan, realized record, and exact conditioned- x grid. It shall expose its coordinate-specific numerical operator, unit/sign/reference, response, support, validity, uncertainty availability, mapping, raw parentage, leakage lineage, and provenance without becoming an independent pipeline or inheriting x -specific correction.
Role	SCI-RTC-REQ-109	Conditioned r shall be optional requested content of the atomic RTC bundle. Its request shall prescribe neither a universal default nor an independent lifecycle; it shall remain uncalibrated, non-Stokes, numerically unable to alter conditioned x , and ineligible for SCI-CAL. Any future downstream $x \leftarrow r$ nuisance prediction remains explicit PTC-owned authority.
Artifacts	SCI-RTC-REQ-110	Artifact detection and affected-support decisions shall be pair-coherent while evidence and correction amplitudes may be coordinate-specific. A pathology affecting either coordinate shall conservatively affect the paired occurrence. Existing x repair need not numerically replace r ; absent a justified r correction, conditioned r and its response shall be typed invalid or unavailable over the affected support without donor reconstruction or numerical coordinate mixing.
Grid	SCI-RTC-REQ-111	When requested, conditioned r shall occupy exactly the conditioned- x grid, cardinality, selected times, phase-zero slots, and representative occurrences. Invalid or unavailable locations shall be retained in place rather than dropped, reindexed, interpolated, or filled. Local or global conditioned- r failure shall never corrupt, change, or invalidate otherwise valid conditioned x .
Response	SCI-RTC-REQ-112	The paired realized ordinary numerical response shall have identical coordinate-diagonal blocks and zero cross branches as in equation (SCI-RTC-EQ-040). Applicable filtering, masks, state, sampling, grid, and pair-level support shall be identical; the x donor exception shall be absent from the r response and shall create typed r unavailability. The bundle shall retain raw mapping, units/sign/reference, source and leakage lineage, response, validity, support, and uncertainty status. If an admitted joint covariance is available, its propagated cross-coordinate blocks shall be preserved as in equation (SCI-RTC-EQ-041); otherwise numerical covariance may remain unavailable.
Leakage	SCI-RTC-REQ-113	Nonzero optical response in r shall remain diagnostic evidence, not pathology or invalidity, and RTC shall not remove, rotate, or renormalize it to an ideal coordinate. Raw and conditioned leakage diagnostics shall remain stage-distinct. Source protection for artifact learning/correction shall be pair-coherent; absent required source authority, the affected learned or corrective operation shall be unavailable, while an already resolved applicable linear filter may still act. RTC reports these facts and does not decide PTC or SCI-VAL policy.
Handoff	SCI-RTC-REQ-114	Downstream consumers shall receive conditioned x , requested conditioned r or its exact local/global availability state, and sufficient RTC mapping, support, response, validity, cause, uncertainty, leakage, source-protection, and provenance facts to know what happened to the raw pair. RTC shall not convert those facts into a PTC joint-mode decision or a cross-package aggregate validity/usage policy.

Area	ID	Normative statement
Identity	SCI-RTC-REQ-115	Canonical pair learning and conditioning shall begin from the exact admitted raw x/r occurrence pair and shall preserve its ordered identity, independent coordinate-local raw validity, mapping revision, units, sign, reference, normalization, Tune/loading state, and immutable parentage.
Operator	SCI-RTC-REQ-116	Every ordinary jointly applicable canonical stage shall use exactly $J_{\text{pair},\Pi} = I_2 \otimes L_{\Pi}$ on $\mathcal{A}_x \cap \mathcal{A}_r$, with coordinate-local projections elsewhere on the unchanged common grid. Both numerical cross-coordinate branches shall remain zero. Any r -specific optimized operator shall be a separately named descendant with new authority, never the canonical companion.
Operator	SCI-RTC-REQ-117	The two canonical coordinate branches shall have identical masks, pair-level action support, segments, transitions, guards, coefficients, state, initialization/reset, filter definitions, order/precision, phase/delay, boundaries, downsampling, selected times, representative occurrences, cardinality, and output grid.
Evidence	SCI-RTC-REQ-118	Learn shall retain E_x , E_r , and genuinely joint E_{xr} as distinct records with coordinate of origin, exact identity, estimator, physical/sample support, source protection, Tune/mapping state, uncertainty, quality, causal class, direct/inferred status, and source/frequency domain where applicable. Later pair action shall not erase or relabel evidence origin.
Union	SCI-RTC-REQ-119	For every owner-designated hard class whose admission predicates pass, Resolve shall form $B_{\text{pair}} = B_x^{\text{hard}} \cup B_r^{\text{hard}} \cup B_{xr}^{\text{hard}}$ and shall apply the same resolved masks, guards, segments, resets, and affected support to both coordinates while preserving every contributing cause.
Semantics	SCI-RTC-REQ-120	RTC shall not infer universal pair contamination from an undifferentiated generic flag. Pair action requires typed accepted direct- x , direct- r , joint, inferred- x -from- r , or inferred- r -from- x cause under the selected class policy; inferred consequence shall not be reported as direct evidence.

Area	ID	Normative statement
Protection	SCI-RTC-REQ-121	Expected astronomical, atmospheric, leakage, Tune, loading, mapping, or source-class behavior in either coordinate shall not be treated as contamination without the required source-protection and causal predicates. Missing required protection authority shall make the affected learned or corrective operation unavailable rather than convert protected or ambiguous evidence into a hard event.
Pair action	SCI-RTC-REQ-122	Accepted hard direct- r or joint evidence shall be permitted to alter the resolved conditioned- x masks, segments, guards, reset boundaries, support, and plan selection through pair-level Resolve, while retaining zero fixed-state numerical $r \rightarrow x$ response.
Pair action	SCI-RTC-REQ-123	Accepted hard direct- x or joint evidence shall analogously determine pair action for requested conditioned r , subject to its coordinate-local numerical correction and availability state; no x value or correction amplitude shall be copied into r .
Lifecycle	SCI-RTC-REQ-124	Resolve shall compose all retained evidence, protection state, class-specific predicates, accepted/rejected/candidate dispositions, and policies into one complete immutable pair plan before Apply. The plan shall explicitly distinguish common ordinary action from every coordinate-specific affine correction or x -only donor exception.
Lifecycle	SCI-RTC-REQ-125	Apply shall neither discover a new event nor adapt thresholds, coefficients, state policy, action support, or any other plan member within one apply pass. New evidence shall enter only through a separately identified successor learn–resolve–apply attempt.
Level shift	SCI-RTC-REQ-126	An accepted paired level shift shall have one event identity, physical transition interval, timing uncertainty, coherence state, guards, segment split, and reset disposition shared by both coordinates, while preserving separate direct/joint evidence and direct/inferred causes.

Area	ID	Normative statement
Level shift	SCI-RTC-REQ-127	Each stable plateau correction shall be coordinate-specific: b^x , b^r , their estimators, reference orientations, uncertainties, quality states, support sufficiency, availability, and numerical actions shall be recorded independently even though the event boundary and ordinary operator are common.
Level shift	SCI-RTC-REQ-128	RTC shall not copy, equate, rotate, regress, or otherwise infer one level-shift amplitude from the other coordinate. Failure of one required coordinate-specific plateau correction shall leave the shared event, grid, and segmentation intact and shall make only that coordinate's value, response, and dependent covariance components unavailable on the affected plateau cells and throughout the full downstream causal support of the selected common operator, including state to the next reset where applicable.
Spectral	SCI-RTC-REQ-129	An r -origin spectral feature shall remain candidate evidence and shall not automatically notch, subtract, or otherwise alter x . Admission of a shared spectral action shall require the owner-selected cross-coordinate predicate and complete x astronomical-transfer, phase, support, covariance, source-protection, and false-detection budgets.
Spectral	SCI-RTC-REQ-130	Once a shared spectral action is admitted into the canonical pair plan, its exact notch or filter operator, coefficients, precision, phase, state, masks, support, guards, response, and sampling shall be identical on x and r under equation (SCI-RTC-EQ-040).
Donor	SCI-RTC-REQ-131	The existing donor recovery shall remain an explicitly typed x -only exception. It shall create no r donor, cross-coordinate donor, copied value, regression value, interpolation, or other numerical reconstruction of conditioned r .
Donor	SCI-RTC-REQ-132	Every x -only donor recovery shall make requested conditioned r , its local response, and dependent covariance components typed unavailable over the complete causal operator influence of that recovery, not merely at the replaced representative occurrence; raw- r parentage and all pair causes shall persist.

Area	ID	Normative statement
Grid	SCI-RTC-REQ-133	Requested conditioned r shall remain on exactly the conditioned- x grid through all local invalid or unavailable intervals, including the x -donor exception. No row may be dropped, reindexed, filled, or assigned a different phase, time, representative occurrence, cardinality, or boundary.
Provenance	SCI-RTC-REQ-134	The canonical output bundle shall include the raw pair, requested conditioned coordinates or exact availability, evidence and origin, accepted/rejected/candidate dispositions, unioned pair support, direct/inferred causes, common operator identity, coordinate-specific affine actions, donor exceptions, mapping and source-protection state, leakage lineage, response, uncertainty, and realized provenance sufficient to reconstruct what happened.
Uncertainty	SCI-RTC-REQ-135	Every paired statistical claim shall distinguish the common grid, pair-action support, coordinate-local numerical availability, joint covariance support, fixed-state conditional covariance, retained input cross blocks, coordinate-specific affine-correction uncertainty, learned-parameter uncertainty, data-dependent selection/plan uncertainty, donor- or correction-induced coordinate unavailability, and total-derivative availability. Unknown or unsupported components shall be unavailable rather than zero.
Naming	SCI-RTC-REQ-136	Product identity and prose shall use “canonical conditioned pair” only where both conditioned coordinates are requested and numerically available on the stated support. Otherwise it shall state conditioned x , optional same-grid conditioned r or its exact availability, and immutable paired raw- r parentage. A future separately optimized r diagnostic descendant shall remain distinct; none shall silently inherit another’s response or qualification.
CAL	SCI-RTC-REQ-137	SCI-CAL shall continue to consume and calibrate conditioned x only. RTC shall not calibrate r , create an r Stokes meaning, or authorize any automatic $r \rightarrow x$ subtraction; any future $x \leftarrow r$ nuisance prediction remains explicit PTC-owned authority.
Handoff	SCI-RTC-REQ-138	RTC shall hand downstream consumers exact pair facts and causes without selecting a PTC joint- r mode, a diagnostic or nuisance-prediction policy, eligibility, named scientific use, or cross-package aggregate validity/usage. PTC and SCI-VAL shall bind those decisions under their own separately authorized contracts.

Area	ID	Normative statement
Mapping	SCI-RTC-REQ-139	Tune/readout authority shall map native measured (I, Q) to the native pair $(x^{\text{acq}}, r^{\text{acq}})$ with exact identity; a separately identified ALIGN relation shall map that native pair to admitted (x^A, r^A) . RTC shall begin at the admitted aligned pair, retain both upstream authorities, and never apply ALIGN a second time.
Replay	SCI-RTC-REQ-140	Every canonical iterative refinement evaluation, accepted-plan replay, restart, and lineage record shall use the original admitted pair $\mathbf{u}^{(0)}$. Conditioned- x -only publication shall not remove r from learning or create an x -only replay parent.
Support	SCI-RTC-REQ-141	RTC shall record $\mathcal{G}_{\text{pair}}$, $\mathcal{A}_{\text{pair}}$, $\mathcal{A}_x$, $\mathcal{A}_r$, and $\mathcal{A}_C$ as distinct states satisfying equation (SCI-RTC-EQ-042). Common pair identity, action support, masks, segments, guards, state, or ordinary response support shall not assert joint numerical or covariance availability.
Influence	SCI-RTC-REQ-142	A required but unavailable coordinate-specific affine correction shall propagate typed unavailability through the exact downstream support of every later common FIR, IIR, notch, state/reset, and sampling operation. The common grid and other coordinate's availability shall remain independently preserved.
Lineage	SCI-RTC-REQ-143	An admitted affine correction to conditioned r shall not overwrite, replace, or reinterpret the immutable raw- r parent or event evidence. The bundle shall retain the original readout step, raw and conditioned stage identities, correction amplitude/action, support, response, uncertainty, and provenance.

6 Falsifiable predictions and limiting cases

Each prediction is an algebraic or semantic falsification target. A future conformance activity may choose an exact numerical tolerance justified by the operation and representation before observing candidate output. No result is reported by this document.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-001	Identity chain: paired raw application context, $A = I$, no replacement or filter, $M = 1$, valid original pairs, and conditioned r requested. Both values, raw coordinate identity, pair order, cardinality, response, flags, and support are unchanged.	Missing/reordered partner, hidden calibration, value/order/cardinality change, flag loss, nonidentity or cross-coordinate kernel, or new claim of timing/coordinate authority.
SCI-RTC-PRED-002	Context neutrality: feed the same paired $\mathbf{x}\mathbf{s}/\mathbf{r}\mathbf{s}$ input and resolved numerical plan to contexts historically labeled Beammap, Pointing, OOF, Science, and diagnostic-only. The same consumer-neutral bundle schema results; labels do not change operation availability or signal meaning, while consumers select the members they require.	A label silently enables, disables, or changes an operation; changes bundle shape or raw meaning; gives r an independent plan; labels RTC output mJy/beam; or interchanges CAL and RTC output.
SCI-RTC-PRED-003	Raw donor direction: choose $\phi_q = 2$, $\phi_d = 4$, and donor $x_q = 3$. Equation (SCI-RTC-EQ-002) predicts raw target replacement 1.5 and equal detector-static comparison coordinates $\chi_q = \chi_d^R = 6$.	Use of ϕ_d/ϕ_q , output other than 1.5, unequal comparison coordinate, calibration claimed from χ , or dependence on legacy responsivity .
SCI-RTC-PRED-004	Raw donor availability: independently make either factor invalid, convention/domain-incompatible, occurrence-mismatched, or set $\phi_d = 0$. The flxscale route is unavailable.	Finite replacement, stale factor reuse, row-number-only binding, zero insertion, or success without separate donor authority.
SCI-RTC-PRED-005	Order: trace a paired raw input through pair-coherent spike/shift decisions, coordinate-specific correction, ordinary coordinate-diagonal temporal stages, and common phase-zero sampling, then trace only conditioned x through distinct downstream SCI-CAL while retaining raw r and requested conditioned r .	Donor replacement precedes resolved boundaries or crosses one, x repair is copied into r , atmospheric diagnostic evidence is subtracted, absolute calibration or target atmosphere appears inside RTC, r enters CAL, or the CAL parent omits the complete RTC plan.
SCI-RTC-PRED-006	Interior target/donor impulses under fixed selection. Each output column equals equation (SCI-RTC-EQ-012); donor energy appears in the declared detector column, support, scale, and phase.	Missing or circularly wrapped energy, unexplained mixing, wrong donor identity/direction, support, normalization, or phase.
SCI-RTC-PRED-007	Constant input with sufficient context. Output equals the constant times realized DC gain after the declared transient; a DC-preserving chain is exactly constant.	Drift, divide-by-zero mask normalization, edge values advertised as interior, or a false DC-preservation claim.
SCI-RTC-PRED-008	Interior step with complete context. Response equals the cumulative impulse response and settles according to DC gain and stability.	Undeclared overshoot, failure to settle, hidden state leakage, or disagreement with the impulse-derived response.
SCI-RTC-PRED-009	Affine ramp through each FIR and admitted alignment mapping. Output slope and offset follow the zeroth and first moments of the exact kernel.	Unexplained clipping, wrong phase/slope/offset, or a ramp-preservation claim without the required moments.
SCI-RTC-PRED-010	Long bin-centered and off-bin sinusoids at DC, passband interior/edge, transition, stopband, and near output Nyquist. Amplitude and phase follow equations (SCI-RTC-EQ-013) and (SCI-RTC-EQ-014).	Gain/phase outside preregistered tolerance, unrecorded frequency convention, or an aliased component labeled passband signal.
SCI-RTC-PRED-011	Sinusoids at the realized notch center and symmetric offsets under the exact rate and state. Depth, width, phase, and transient follow realized zeros/poles.	Center/depth/width mismatch, unstable pole, rate mismatch, hidden state, or an edge transient reported ordinary-valid.
SCI-RTC-PRED-012	Coordinate mask with valid inside/outside values plus invalid, unavailable, wrong-frame, out-of-domain, and ambiguously bound coordinates. Invalid classes fail the affected operation and remain distinct from outside.	Invalid becoming outside/unmasked, entering geometry, clamping to a center/edge, or loss of the distinct cause.
SCI-RTC-PRED-013	Samples just inside, exactly on, and just outside each selected mask boundary and temporal dilation. Membership follows the recorded inclusive/exclusive and dilation policy identically across execution modes.	An undocumented boundary, frame mismatch, asymmetric dilation, or sequence-dependent membership.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-014	Impulses and constants at first/last support cells and immediately inside complete FIR context. Values, guard, support, and output admission follow the selected edge rule.	Undeclared padding, truncation, wrap, renormalization, guard, or output selection.
SCI-RTC-PRED-015	Zero, impulse, and constant inputs split and rejoined at scan/chunk/state boundaries for an IIR. Results follow the declared reset/carry and terminal-state policy.	Prior-observation leakage, hidden carry/reset, unbounded non-finite poisoning, or contradiction between split and declared state semantics.
SCI-RTC-PRED-016	Lengths $0, 1, L - 1, L, L + 1$ around every support/guard threshold, with stable scan identity.	Crash, underflow, stale output, silent scan deletion/renumbering, or ordinary-valid status without required context.
SCI-RTC-PRED-017	Inject NaN, $+\infty$, and $-\infty$ separately into a sample, factor, coordinate, coefficient, and IIR state. Failure or unavailable influence follows REQ-026 exactly.	Silent zero/coercion, finite-looking output without cause, stale state recovery, or an unbounded tail reported valid.
SCI-RTC-PRED-018	Place synthesized and replaced causes first at the directly selected occurrence and then at noncenter members of full support. Direct cases are universally excluded; noncenter causes propagate unchanged and consumer eligibility follows the declared policy.	A direct case called independent, a noncenter cause erased or made universally ineligible by RTC, or consumer disposition reported without its policy.
SCI-RTC-PRED-019	For $M = 1, 2, 3, 4, 5$ and $N = 0$ through several incomplete final strides, enumerate $j = Mn$. Cardinality and selected time follow equation (SCI-RTC-EQ-011); no block mean appears.	Wrong N_y , dropped/duplicated point, nonzero phase, averaged value, timestamp misidentification, or rate mismatch.
SCI-RTC-PRED-020	Two sinusoids separated by an integer multiple of f_{out} , before and after the realized prefilter. Folded amplitude is the sum in equation (SCI-RTC-EQ-014).	A folded term missing from response, inadequate stopband labeled alias-free, or incorrect image frequency/amplitude.
SCI-RTC-PRED-021	Reuse one donor for two targets and give neighboring outputs overlapping filter support. Any covariance claim states whether the resulting cross-target and temporal correlations are included, excluded, or unavailable; when included they follow equation (SCI-RTC-EQ-016b) and the admitted covariance.	Undisclosed omission, excluded or unknown correlations treated as zero, independent weights asserted without evidence, or reused donor described as additional exposure.
SCI-RTC-PRED-022	Enable a response-changing mask, donor mix, stateful filter, edge, and decimator while providing only a scalar FIR transfer. Complete response state is unavailable.	The partial scalar transfer labeled complete, or downstream use that assumes omitted stages are identity.
SCI-RTC-PRED-023	Learned candidates include at least two safe factors and one failing each astronomical-transfer, alias, sampling, realizability, and compatibility constraint. Equation (SCI-RTC-EQ-021) chooses the largest passing factor.	A smaller safe factor selected without rule, any failing factor applied, a percentile speed used for safety, or a missing numeric predicate treated as pass.
SCI-RTC-PRED-024	Serialize and restart a learned plan, then alter one of beam identity, speed support, scan set, coefficient precision, rate, phase, application context, consumer binding, or parent. Apply rejects the mismatch.	Silent relearning, retuning, native fallback, stale-plan application, or a realized record rewriting the application context or plan.
SCI-RTC-PRED-025	Execute two observations sequentially with distinct rates, constants, factors, masks, donors, states, and output parents.	Any coefficient, state, factor, donor, mask, rate, flag, or product link leaks from the first observation.
SCI-RTC-PRED-026	Toggle the complete spike-event/donor manifest and other optional inert diagnostics while holding the compact treatment/population summary, resolved plan, and realized operator fixed. Values, flags, influence, eligibility, cardinality, response, failures, and normal summary are identical.	Despiking ceases to modify accepted targets, any scientific or numerical result changes, or an observational/production claim appears from the detail toggle.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-027	Bind identical learned parameters first to a separate training population and then to the apply observation. The numerical fixed-state response can match, but the second plan records data dependence and parameter/selection uncertainty.	Same-observation learning represented as fixed a priori, or conditional linearity used to erase selection or parameter uncertainty.
SCI-RTC-PRED-028	Remove in turn one required learning-population identity, tolerance, uncertainty, or downstream-compatibility predicate before resolve. The dependent plan remains unresolved and apply does not begin.	Missing evidence treated as pass, a default inserted, or output represented as applied learned mode.
SCI-RTC-PRED-029	Serialize one resolved plan and attempt during apply to alter a notch center, cutoff, tap count, donor rule, or decimation factor. The mutation is rejected or belongs to a separately authorized adaptive estimator.	Hidden retuning, silent fallback, or a realized record that rewrites the resolved plan.
SCI-RTC-PRED-030	Sweep notch center error, line drift, width, depth, Q/order, initialization, direction, and overlap. Residual line and astronomical amplitude, phase, centroid, covariance, transient, and edge loss follow the exact combined response.	Only line disappearance measured, or any changed astronomical observable omitted from the plan's validity evidence.
SCI-RTC-PRED-031	Scan the same projected source at multiple speeds and directions through a fixed filter. Its temporal spectrum and retained amplitude/PSF change according to equation (SCI-RTC-EQ-023) and exact response.	One nominal cutoff or scalar beam treated as speed- and direction-independent proof of preservation.
SCI-RTC-PRED-032	Design FIR candidates over multiple tap counts and coefficient precisions. The selected N_* is the smallest candidate passing every registered transfer, alias, phase, support, guard, and realizability predicate.	Arbitrary configured taps, approximate order formula promoted to authority, or a candidate with any failed predicate selected.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-033	Inject compact and extended sources through admitted high- or band-pass candidates. Low-frequency rejection produces the predicted lost broad modes, compensated response, or negative lobes.	Unit preservation labeled flux preservation, or removed absolute/extended modes reported available without response restoration evidence.
SCI-RTC-PRED-034	Choose donor and target with compatible flxscale but deliberately different sky signal or beam coupling. Raw-domain convention transfer remains algebraically correct while the surrogate is not a missing-sky reconstruction.	Compatible factors claimed to prove physical donor equivalence or independent exposure.
SCI-RTC-PRED-035	Attempt Beammap donor scaling with a new flxscale derived from that same conditioned Beammap. The route is rejected until an identified prior factor authority is supplied.	Circular use of the new factor or absence of the prior authority in lineage.
SCI-RTC-PRED-036	Apply filters with known nonzero group delay to Pointing/Beammap scans in opposite directions. Centroid displacement follows equation (SCI-RTC-EQ-027) with opposite scan-direction sign unless time/coordinates are corrected.	Selected grid time treated as response center, wrong displacement sign, or unrecorded correction.
SCI-RTC-PRED-037	Process calibrator and target with plans that share a nominal label but differ in coefficients, rate, phase, state, edge, or decimation. Cross-plan calibration transfer is unavailable absent explicit response accounting or observational evidence.	Nominal label, preserved unit, or flxscale alone treated as RTC-plan compatibility.
SCI-RTC-PRED-038	Pass all deterministic coefficient and operator fixtures but withhold the required source/scan design studies. Algebraic correctness may be supported; science qualification remains unassessed.	Deterministic pass promoted to observational performance, science qualification, or production readiness.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-039	Inject one strong coherent line and no other valid line. Attempt 1 accepts and suppresses it; Attempt 2 finds no acceptable successor. The accepted-plan count remains one and Π_1 is final with a no-successor stop.	A manufactured Π_2 or evaluation product, unbounded iteration, unsupported second notch, rejected evidence lost, or no typed stop/final-plan identity.
SCI-RTC-PRED-040	Inject two unequal coherent lines with the weaker revealed only after the stronger is suppressed, including decisive r -origin evidence. A later attempt accepts one complete two-notch pair plan as Π_k and replays it once on original $\mathbf{u}^{(0)}$; projected outputs follow the predicted cumulative paired response.	x -only replay parent, r evidence lost, attempt and accepted-plan indices conflated, increment-only plan, accumulated exploratory parent, missing first notch, duplicate filtering, or response disagreement.
SCI-RTC-PRED-041	Create a peak at a first-notch edge that agrees with predicted cumulative ringing or response artifact. The successor candidate is rejected and the first plan remains final.	Artifact admitted as a line, response floor ignored, or prior accepted plan invalidated solely by rejection.
SCI-RTC-PRED-042	Compare applying a one-notch pair plan twice with applying the intended complete successor once to original $\mathbf{u}^{(0)}$. Values, paired response, phase, edges, support, availability, and covariance differ unless exact operator equivalence is proved.	Repeated filtering treated as original-pair replay, an x -only parent used, or equality claimed from nominal notch identity without complete operator proof.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-043	Propose a successor that fails one cumulative transfer, stability, support, or compatibility predicate. The attempt is rejected; its evidence and cause persist while the accepted-plan index and final product remain unchanged.	Rejected proposal assigned a new accepted-plan index or evaluation, failed plan applied, prior product rewritten, rejected evidence erased, or missing predicate treated as pass.
SCI-RTC-PRED-044	Alternate between proposing addition and removal of the same line or vary its parameters outside declared stability across attempts. Termination is typed oscillating or nonconvergent.	Alternation or parameter instability labeled convergence, stable plan, or successful maximum-safe refinement.
SCI-RTC-PRED-045	Supply a sequence that reaches the owner-approved maximum attempt count without satisfying plan stability. The result records maximum-attempt termination and the last accepted plan, not convergence.	Maximum count labeled stable/converged, continued iteration, attempt count confused with accepted-plan count, or disappearance of the last accepted plan.
SCI-RTC-PRED-046	Serialize and restart the final cumulative plan, then vary original parent or learning-population identity. Exact compatible restart reproduces the product; either mismatch is rejected.	Silent relearning, stale-parent application, altered plan, or acceptance of incompatible learning lineage.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-047	Supply complete paired occurrences, then remove only one r partner and separately only one x partner. Complete pairs are admitted; either incomplete occurrence fails required-input admission.	Partner synthesis, copying, zero fill, independent join, or successful admission of an incomplete pair.
SCI-RTC-PRED-048	Apply a known native nonlinear $IQ \rightarrow xr$ mapping, then one admitted ALIGN relation, and serialize/restart. The aligned pair and both upstream identities reproduce equations (SCI-RTC-EQ-030) and (SCI-RTC-EQ-004); the RTC-local operator begins at that aligned pair.	A nonlinear mapping is forced into one global affine form, either upstream identity changes, or ALIGN is omitted or applied twice inside RTC.
SCI-RTC-PRED-049	Make x numerically invalid and r valid, then reverse them, while preserving a pair-level pathology cause/support. The pair persists and coordinate availability remains distinct; invalid r does not alter valid x .	One member's numerical state copied to the other, invalid member used as valid, pair-level cause/support lost, or valid x altered because r is unavailable.
SCI-RTC-PRED-050	Inject an optical signal with known nonzero r/x response, then rescale the r coordinate under an explicit mapping revision. Each estimator recovers the correspondingly transformed ϵ ; a reported ψ is invariant only after the declared normalization and metric transform.	r assumed zero, a ratio or angle remains numerically unchanged under incompatible rescaling, or units, mapping revision, metric, uncertainty, or support are missing.
SCI-RTC-PRED-051	Give atmosphere and bright source deliberately different leakage ratios and compare them first under compatible and then incompatible coordinate conventions. Distinct products remain available; the incompatible comparison is unavailable until an explicit transform is applied.	Averaging, substitution, universal promotion, or cross-revision comparison without a compatible normalization and metric.
SCI-RTC-PRED-052	Generate frequency-dependent leakage and fit both a scalar and admitted spectral estimator. Only the spectral product supports the frequency-dependent claim; the scalar is range-qualified.	Scalar advertised as broadband authority, missing band/support, or absent response status.
SCI-RTC-PRED-053	Attempt to correct x from r , correct r from x , calibrate r , use either coordinate as a donor for the other, or replace either invalid coordinate with its valid partner. Every route is unavailable in v0.1.	Any route executes or is implied by shared pair identity.
SCI-RTC-PRED-054	Simulate correlated x/r noise and an r -dependent selector with both conditioned coordinates requested. At fixed state, the numerical Jacobian is coordinate-diagonal; the joint conditional covariance retains propagated cross blocks. Across selector realizations, selection uncertainty is retained or unavailable.	A cross-coordinate numerical branch appears, admitted cross blocks are discarded or forced to zero, the conditioned- x upper-left block changes because r fails, or selector dependence is discarded.
SCI-RTC-PRED-055	Add one network event with a known finite physical-time transition, bounded detector timing offsets, unequal x/r evidence, and adjacent isolated spikes; repeat it at different sampling cadences. Learning uses the original pair, reports one event identity, cadence-consistent timing-vector-derived transition support, stable plateaus, and only then permits valid additive correction and within-segment replacement.	A fixed sample count changes the inferred physical width, detector events lose their network parent, donor-repaired values drive the change point, shift is absorbed as spikes, amplitude is copied between coordinates, or a donor crosses the boundary.
SCI-RTC-PRED-056	Place isolated spikes immediately before and after a level shift and vary donor values independently. The resolved change point is invariant to unused donor values; replacement begins only after segmentation and stays inside each stable segment.	Change-point learning consumes donor substitutions, depends on donor policy, crosses a shift boundary, or circularly repairs evidence used to resolve that boundary.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-057	Put a filter impulse on each side of a declared level-shift boundary. The ordinary plan resets. A carry plan is admitted only with separately authorized continuity, complete response, and a passing residual-contamination criterion.	Silent cross-boundary state, ordinary carry without separate authority, failed/missing residual criterion, ordinary-interior labels in guards, or missing reset provenance.
SCI-RTC-PRED-058	Insert two level shifts with overlapping guards and a short middle segment. Event identities remain ordered and the selected conflict/minimum-segment policy gives one deterministic disposition.	Event collapse, order dependence outside policy, negative/ambiguous segment, or silent guard truncation.
SCI-RTC-PRED-059	Give stable pre/post plateaus a known additive offset around a finite transition. With correction disabled the values are unchanged; with a valid selected correction one plateau moves to the declared reference while transition cells remain flagged, unmodeled, and excluded.	An unselected correction changes values, the reference/sign is ambiguous, transition cells enter either plateau fit or become valid, the cause is erased, or a gain term is fitted.
SCI-RTC-PRED-060	Measure leakage before and after declared coordinate-diagonal filtering. The raw and conditioned diagnostics differ according to their composed responses and retain separate identities, units, mappings, support, and uncertainty.	One value reused across stages, nonzero r optical response removed or renormalized, filtering treated as commuting without proof, or pre/post parentage lost.
SCI-RTC-PRED-061	Hold the resolved x operator fixed and toggle an atmospheric leakage template between available and diagnostic-unavailable states. Conditioned x is unchanged while diagnostic availability and lineage change exactly as declared.	The template is numerically subtracted, changes x , silently becomes PTC cleaning, or is conflated with target SCI-CAL atmosphere.
SCI-RTC-PRED-062	Stop iterative refinement before $A_{\max}$ after A actual attempts. The record satisfies $K \leq A < A_{\max}$ and contains no fabricated attempts.	$A = A_{\max}$ forced, artificial no-op attempts, or accepted-plan count confused with either attempt quantity.
SCI-RTC-PRED-063	For Beammap, Pointing, OOF, Science, and diagnostic-only application contexts, inspect resolved plans and output bundles. The consumer-neutral schema retains conditioned x , requested conditioned r or typed unavailability, raw r parentage, paired mapping/validity, leakage, level-shift/segment/reset, diagnostic atmosphere treatment, and consumer routing.	Missing disposition, context-dependent loss or shape change, an independent r lifecycle, x correction silently applied to r , an operation enabled by label implication, or consumer routing that bypasses downstream contracts.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-064	Apply an ideal admitted mapping to a purely optical excitation whose nominal orthogonal projection is zero. The recovered r optical coefficient is consistent with zero within the preregistered estimator tolerance.	A nonzero coefficient outside tolerance, missing uncertainty, or a zero claim made without adequate excitation/support.
SCI-RTC-PRED-065	Apply a known small mapping rotation to the same excitation. The measured r/x leakage follows the predicted sign and magnitude over the declared local domain.	Wrong sign/magnitude, global-affine extrapolation outside the mapping domain, or rotation absorbed into an unrelated nuisance.
SCI-RTC-PRED-066	Generate shared optical signal with noise in both x and r , then compare direct r -on- x regression with the admitted estimator. Direct regression exhibits the predicted errors-in-variables or shared-data bias unless its method explicitly accounts for it.	Raw correlation promoted to an unbiased coefficient, self-inclusion ignored, or estimator uncertainty excludes shared-data dependence.
SCI-RTC-PRED-067	Scan an optical source whose expected r leakage follows the admitted source response across a bounded interval. The crossing remains a source/leakage event rather than an abrupt persistent level shift.	Expected optical leakage alone triggers a shift, creates a false plateau boundary, or loses source identity.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-068	Inject a gradual drift over declared support. It remains drift/nonstationarity evidence unless the selected change-point model independently supports an abrupt transition.	The drift is automatically collapsed into one step or an abrupt-shift label is assigned without model evidence.
SCI-RTC-PRED-069	Learn a PSD/notch plan first from data containing an unsegmented step and then from the same original data with the accepted transition and guards excluded. The first plan shows the predicted step contamination; the second excludes it under the declared support policy.	Identical stationary-learning claims without accounting, a step learned as a spectral line, or segmentation that still admits cross-boundary support.
SCI-RTC-PRED-070	Give stable pre/post plateaus materially different admitted leakage or paired optical-response states. The event remains distinct, additive correction is unavailable, and v0.1 fits no gain or responsivity change.	Additive equivalence is claimed despite the response change, a gain-changing correction is silently fitted, or pre/post comparison is omitted when support exists.
SCI-RTC-PRED-071	Construct one short plateau that supports readout-health diagnostics but not a defensible offset or stronger requested operations, alongside another usable plateau. The event remains a boundary, no offset is invented, and the plan separately retains or rejects each plateau for the application context.	One sample-count threshold controls all uses, an offset is fabricated, the entire scan is automatically discarded, or diagnostic validity is promoted to a stronger use.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-072	Toggle the conditioned- r request while holding admitted input and the conditioned- x plan fixed. Conditioned x is bitwise/numerically identical, raw- r parentage remains, and the request changes only optional paired content and its facts.	Conditioned x , its plan, response, support, or CAL eligibility changes; r becomes calibrated/Stokes; or a separate r lifecycle appears.
SCI-RTC-PRED-073	Put coordinate-specific spike evidence in only r and select an x -donor repair over the resulting paired support. The pathology/support affects the paired occurrence, x follows its justified repair, and conditioned r remains present but typed invalid/unavailable without donor reconstruction.	The pathology is ignored in x , r is donor-filled or copied from x , coordinates are numerically mixed, or coordinate-specific evidence is lost.
SCI-RTC-PRED-074	Request conditioned r with several local invalid intervals and one globally unavailable r operator state. Its grid, cardinality, selected times, and representative occurrences exactly match conditioned x ; invalid locations remain in place, and valid x is unchanged in both cases.	Rows are dropped/reindexed/interpolated/filled, grids differ, or local/global r failure corrupts or invalidates valid x .
SCI-RTC-PRED-075	Apply a shared filter and phase-zero selector to correlated paired input while adding one x -only donor repair. Ordinary responses use identical diagonal L_{Π} blocks, common grid/support agree, the donor remains an explicit x exception with typed r unavailability over its influence, and available joint covariance follows equation (SCI-RTC-EQ-041).	Either cross numerical derivative is nonzero, ordinary diagonal blocks differ, a donor value appears in r , available cross-covariance is erased, or an absent covariance model is reported as numerical zero.
SCI-RTC-PRED-076	Inject an expected optical source with nonzero r response into artifact-learning support. With required source authority present, the source is pair-protected and raw/conditioned leakage remains separately characterized; with that authority absent, the affected learned/corrective operation is unavailable while an already resolved applicable linear filter remains coordinate-diagonal.	Expected leakage alone is flagged pathological, removed/rotated/renormalized, used as artifact evidence without protection, raw/conditioned stages conflated, or a resolved filter is recast as an invented correction.
SCI-RTC-PRED-077	Deliver one bundle with valid conditioned x , locally unavailable conditioned r , and complete RTC cause/support/response/leakage/source-protection facts to two consumers with different declared policies. RTC content is identical; consumer dispositions may differ under their own authority.	RTC changes x , selects a PTC joint mode, publishes a cross-package aggregate validity decision, erases causes, or tailors producer facts to a consumer's policy.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-078	Apply an admitted FIR stage to paired impulses with conditioned r requested. Both diagonal blocks reproduce the same coefficients, support, precision, phase, and boundary response.	Unequal impulse responses, coefficients, support, phase, precision, or a nonzero cross-coordinate numerical branch.
SCI-RTC-PRED-079	Apply an admitted stateful filter across identical pair-plan gaps, level shifts, guards, and restarts. Both coordinates use the same state policy, initialization, reset boundaries, and transient classification.	Coordinate-specific ordinary state/reset behavior, silent carry, or different guard/interior classification.
SCI-RTC-PRED-080	Downsample an odd-length paired segment with factor M and several invalid cells. Both products have identical phase-zero slots, selected times, representative occurrences, cardinality, masks, and unchanged rows.	Different phase/cardinality/grid, row deletion, fill, reindexing, or unequal representative occurrence.
SCI-RTC-PRED-081	Hold Π fixed and perturb only r , then only x , on common available support. The opposite numerical output is unchanged and the two self-responses equal L_Π .	Either fixed-state cross derivative is nonzero or the two self-response operators differ.
SCI-RTC-PRED-082	Insert one accepted hard spike seen directly only in r . Resolve retains direct- r evidence, adds inferred- x pair consequence, and uses one unioned action support for both.	x support unchanged, r origin erased, inferred x reported as direct, or evidence values unioned into a generic flag.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-083	Repeat PRED-082 with direct evidence only in x , then with a genuinely joint estimator. The same pair-action rule applies while the three origins remain distinguishable.	Asymmetric hard-action support, loss of origin, or joint evidence represented as two direct detections.
SCI-RTC-PRED-084	Overlap accepted hard x , r , and joint events with different supports. The applied support equals the set union and every contributing cause remains queryable at overlaps.	Intersection, last-writer support, duplicate action, lost cause, or action outside the resolved union.
SCI-RTC-PRED-085	Present a weak r line, leakage-like coherence, and a hard non-finite event. Only the hard accepted event enters B_{pair} until separate candidate predicates pass.	All diagnostics unioned into hard action or the accepted hard cause left out.
SCI-RTC-PRED-086	Hold the resolved plan fixed while new evidence appears during Apply. Current output follows the original plan; a changed plan appears only in a successor attempt.	Within-apply event discovery, coefficient/support mutation, or realized record rewriting Π .
SCI-RTC-PRED-087	Inject expected optical response with nonzero r/x leakage and complete source authority. It remains protected diagnostic evidence while any independently accepted ordinary filter acts identically on both coordinates.	Leakage becomes automatic hard pathology, is removed/rotated, or causes unequal ordinary operators.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-088	Repeat PRED-087 without required source-protection authority. The affected learning/correction is unavailable; expected response is not silently promoted to hard evidence.	Missing authority treated as contamination, pass, or zero leakage.
SCI-RTC-PRED-089	Place an optical source over a real hard discontinuity supported independently of source response. Protection preserves the optical evidence while the typed hard cause still produces pair action.	Source mask suppresses all hard evidence, or hard action erases the protected source/leakage record.
SCI-RTC-PRED-090	Inject one accepted paired level shift with unequal known Δx and Δr . One transition, guard, segment split, and reset are recorded, with independently estimated amplitudes and uncertainties.	Two unrelated event boundaries, copied/equal amplitudes, or a missing coordinate-specific uncertainty.
SCI-RTC-PRED-091	Make the x plateau correction valid and the r correction unsupported, then apply a finite FIR and phase-zero selector. Shared segmentation/grid remain; x may be corrected, while conditioned r is unavailable over the full predicted downstream footprint rather than reconstructed.	Unavailability ends at plateau cells, event/grid changes, x is invalidated solely by r failure, or an r correction is copied from x .
SCI-RTC-PRED-092	Reverse correction availability in PRED-091 and apply an IIR with a known next reset. The shared event persists, valid r correction may act, and unavailable conditioned x influence extends exactly to the declared reset without $r \rightarrow x$ amplitude mixing.	Influence stops early or crosses the reset, r correction numerically changes x , event/grid splits, or missing x correction is invented.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-093	Reverse the selected plateau reference while preserving evidence and event support. Each available coordinate correction changes sign consistently with its own amplitude; transition cells and reset remain identical.	Unchanged sign, copied cross-coordinate amplitude, transition repair, or changed segmentation.
SCI-RTC-PRED-094	Repeat the same physical transition at two admitted cadences. Physical-time event/support and pair identity agree, while sample support follows each timing vector and both coordinates share it per realization.	Fixed sample width, coordinate-specific boundary, or cadence-dependent physical event identity.
SCI-RTC-PRED-095	Place unequal coordinate noise and quality around a shared shift. Event acceptance follows the resolved evidence policy; amplitude uncertainty and correction availability remain coordinate-specific.	One coordinate's quality copied to the other or a valid pair event equated with two valid corrections.
SCI-RTC-PRED-096	Put a level shift adjacent to a stateful-filter guard with one unavailable coordinate correction. The common operator resets and propagates one identical guard/support rule, while coordinate-local numerical availability differs over the exact downstream influence.	Different reset/guard support, pair identity equated with joint availability, unavailable influence omitted, or affine correction folded into ordinary response.
SCI-RTC-PRED-097	Inject an r -only spectral candidate with no passing cross-coordinate predicate. The evidence is retained and neither coordinate receives a new notch.	Candidate automatically notches x or r , or disappears because no notch was admitted.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-098	Supply a spectral candidate that passes every selected cross-coordinate, transfer, phase, support, covariance, protection, and false-detection predicate. One admitted notch with identical response acts on both coordinates.	Notch applied to only one canonical branch, unequal parameters, or admission with a missing/failed budget.
SCI-RTC-PRED-099	Vary candidate strength around the selected shared-admission boundary. Candidate disposition changes under the preregistered rule; ordinary paired action changes only after Resolve.	Hidden threshold, within-apply spectral adaptation, or automatic union of candidate subtraction.
SCI-RTC-PRED-100	Perform one accepted x donor replacement and propagate it through a finite-support filter. Conditioned r is unavailable over the full transitive output influence while its grid and raw parent remain intact.	Only the representative cell is unavailable, r is donor-filled, rows are dropped, or a cause is lost.
SCI-RTC-PRED-101	Reuse a donor across multiple targets with correlated paired input. x records donor-induced correlations; r remains unreconstructed and its affected covariance components are unavailable rather than zero.	Independent-exposure claim, missing donor correlation, invented r covariance, or cross-coordinate donor value.
SCI-RTC-PRED-102	Toggle the x donor exception while holding common ordinary L_{Π} fixed. Outside its causal influence both canonical branches retain identical ordinary response; inside, valid x may remain while r response is typed unavailable.	Donor branch inserted into L_r , ordinary filters diverge elsewhere, or x is invalidated solely by r unavailability.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-103	On a preregistered artifact-free population containing expected r optical response, compare x data loss and false pair actions with and without accepted r -origin hard-evidence pathways, holding all owner-selected predicates fixed. The study reports the trade space and uncertainty without asserting a result in this contract.	No false-positive/data-loss accounting, optical response counted as known contamination, thresholds changed after observing outcomes, or a performance/qualification claim inferred from this prediction.
SCI-RTC-PRED-104	Trace one native occurrence from measured (I, Q) through Tune mapping to $(x^{\text{acq}}, r^{\text{acq}})$, then through exactly one ALIGN relation to (x^A, r^A) and RTC.	Tune output mislabeled aligned, ALIGN omitted or applied twice, identities collapsed, or RTC begins from native rows.
SCI-RTC-PRED-105	Hold the original admitted pair fixed, let decisive evidence occur only in r , and toggle publication of conditioned r . Every iterative evaluation and final replay uses the same $\mathbf{u}^{(0)}$ and pair plan; only the optional output projection changes.	x -only learning/replay, loss of r evidence, changed accepted plan from the publication toggle, or unnecessary conditioned- r artifact required.
SCI-RTC-PRED-106	Create equal pair-action support with valid conditioned x , locally unavailable conditioned r , and joint covariance available only on their overlap. The grid and action support match, $\mathcal{A}_x \neq \mathcal{A}_r$, and $\mathcal{A}_C \subseteq \mathcal{A}_x \cap \mathcal{A}_r$.	Common support treated as complete paired values, rows dropped, covariance reported outside joint support, or one coordinate's availability copied to the other.
SCI-RTC-PRED-107	Make one affine correction unavailable immediately before FIR, IIR, notch, and decimation cases with known support/reset. The affected coordinate's unavailable output follows each exact transitive footprint; the grid and other coordinate remain intact.	Unavailability confined to input cells, wrong finite footprint, IIR influence crossing or missing its reset, row loss, or other-coordinate corruption.
SCI-RTC-PRED-108	Apply a valid additive correction to conditioned r across a known readout step. The corrected companion changes as authorized, while immutable raw- r parentage and event diagnostics reproduce the original step and retain distinct stage identity.	Raw parent overwritten, original step erased from diagnostics, corrected and raw stages conflated, or correction provenance absent.

6.1 Analytic limits

The following limits are required: a compatible donor pair with $\phi_q = \phi_d$ gives unit raw donor scale; a replacement indicator tending to zero gives identity; filter coefficients tending to a Kronecker delta give identity; $M \rightarrow 1$ gives no rate change; a source-mask radius tending to zero selects only exact valid coincidences under the declared boundary; complete context approaches the declared interior response; and a causal pole approaching the unit circle exposes increasing memory and never retains a stability claim at or outside the circle.

End of shared normative core: SCI-RTC-v0.1/r0.12. View-specific material cannot add to or modify this scientific authority.

A Informative conformance-record schema

This appendix supplies bookkeeping guidance only. It does not add a requirement. A review record should contain the following fields.

Field	Evidence content
Record identity	Stable review ID, date, reviewer role, exact contract revision, and supersession status.
Claim layer	Exactly one primary layer: implementation conformity, representation fidelity, observational performance, science impact, or production readiness.
Target identity	Exact source/build/artifact identity and any configuration or runtime identity needed to reproduce the observation.
Authority binding	One or more SCI-RTC requirement or prediction IDs; an equation/definition may be cited only through the shared core.
Preconditions	RTC application context, signal domain, detector occurrences, grid, intervals, immutable resolved plan, realized-record scope, and owner decisions required by the check.
Method	Static trace, deterministic vector, bounded matrix calculation, serialization inspection, failure injection, or other named method.
Expected result	Exact invariant and a tolerance justified before observing candidate output.
Observed result	Values, states, messages, artifacts, and exceptions without interpretation hidden in a generic pass flag.
Disposition	Pass, fail, not exercised, blocked by open owner decision, or evidence unavailable.
Claim limit	Explicit statement of what the record does not establish.

No status may be inferred from an omitted record. “Not exercised” means the evidence was not attempted; “blocked” means a named unresolved authority prevents a meaningful check; and “unavailable” means the attempted evidence cannot support the claim. These dispositions are not passes.

B Informative requirement-review routing

The table suggests evidence classes without prescribing implementation structure. Reviewers should use the exact crosswalk shipped with the package for one-to-one coverage.

Requirement range	Primary review surface	Useful evidence classes
REQ-001–005	Paired raw boundary and CAL handoff	Exact paired input, conditioned- x /requested-conditioned- r /raw- r -parent bundle inspection, x -only downstream CAL-parent failure injection, and atmospheric-diagnostic no-subtraction trace.
REQ-006–012	Identity, context–plan–record state, and selected policy	Serialization round trip, distinct-identity fixtures, one-way context/plan/record trace, context-label neutrality, unsupported-policy rejection.
REQ-013–018	Ordering, despiking, and donor transfer	Raw-RTC-then-CAL order trace, accepted-target modification check, compact spike-population summary, forced donor vectors, invalid-factor matrix, donor reuse/tie/fallback fixtures, and optional detailed-manifest toggle.
REQ-019–020	Direct exclusion and influence	Direct/noncenter synthesis/replacement injection, full-support cause propagation, downstream policy handoff inspection.
REQ-021–027	Filters, masks, edges, and finite policy	Coordinate-applicability and coefficient/state serialization, paired impulse/step/constant/ramp/notch vectors, mask-boundary and non-finite failure injection, and local conditioned- r unavailability isolation.
REQ-028–031	Fixed phase-zero sampling	Enumerated paired length/factor cases, exact x/r grid/cardinality/selected-index/time identity, retained invalid slots, rate trace, and exact folded-band calculation.
REQ-032–036	Learned plan lifecycle	Candidate decision table, maximum-safe selection, common-plan identity, immutable apply and restart mismatch fixtures.
REQ-037–041	Complete response and support	RTC-local coordinate-diagonal factorization from the aligned parent, zero cross-coordinate numerical branches, optional explicit ALIGN composition, local Jacobian fixtures, LTI-domain proof, edge/state/support expansion.
REQ-042–045	Statistical claim disclosure	Conditioned- x fixed-state covariance and paired block-covariance fixtures; for every conditional, component-limited, or total claim, inspect explicit included/excluded components and correlations plus unavailable-state preservation.
REQ-046–051	Validity and consumer-neutral atomic output	Orthogonal-state fixtures, common bundle-schema inspection across contexts, atomic write/failure injection, compact provenance reconstruction, inert detailed spike/donor-manifest toggle.
REQ-052–054	Consumers and claims	Parent-binding inspection, unavailable-state preservation, disabled-PTC terminal path, claim-label audit.
REQ-055–058	Learn–resolve–apply lifecycle	Learning-population identity, predicate-completeness matrix, immutable-plan trace, mutation rejection, adaptive-estimator separation.
REQ-059–063	Scientific filter design	Purpose/model register, scan–beam transfer studies, notch and band specifications, constrained tap/order selection.
REQ-064–068	Donor, decimation, timing, calibration	Donor-meaning separation, circular-factor rejection, folded-band calculation, centroid/coordinate tests, cross-plan response comparison.
REQ-069–070	Validation and claims	Preregistered design sweeps and strict claim-layer disposition.
REQ-071–076	Refinement attempts and replay	Distinct attempt/accepted-plan trace, immutable apply, complete-plan reconstruction, original-input replay, explicit-cascade comparison, intended-consequence assessment.
REQ-077–082	Candidate, stability, stopping, provenance	Artifact discrimination, cumulative budgets, complete-plan stability, actual/max-attempt distinction, typed disposition, lineage, restart mismatch fixtures.
REQ-083–086	Pair admission, identity, mapping, validity	Missing-partner failure, mapping reconstruction, reorder/rejoin rejection, coordinate-specific numerical-validity and pair-level cause/support matrix, restart identity.
REQ-087–093	Leakage, prohibited correction, joint statistics	Atmosphere/source estimator fixtures, coordinate rescaling and comparison compatibility, scalar/frequency status, forbidden bidirectional-route rejection, fixed-state zero cross branches, and joint selector covariance.
REQ-094–102	Level shifts, plateaus, and state	Original-pair spike masking/downweighting; network/detector finite physical-time transition support; timing-vector-derived sample masks; additive-only plateau model and optional valid offset correction; transition exclusion; post-segmentation donor fixtures; reset-by-default boundaries; separately authorized carry; compact production state; and typed lifecycle states.
REQ-103–105	CAL, atmosphere diagnostics, context and consumer routing	x -only CAL handoff, atmospheric-template no-subtraction checks, retained raw- r parentage, consumer-neutral bundle binding, all-class admission, explicit-selection execution, and context-label neutrality checks.
REQ-106–108	Plateau support, pre/post leakage, optional conditioned r	Application-context support matrix, insufficient-offset-support behavior, pre/post response comparison/unavailability, gain-model exclusion, and paired-companion operator completeness.
REQ-109–114	Conditioned- r reopening decisions	Request-toggle x invariance, pair-coherent artifact support without donor r repair, exact paired-grid/local-global failure isolation, coordinate-diagonal response and joint covariance, leakage/source protection, and consumer-policy handoff.

Requirement range	Primary review surface	Useful evidence classes
REQ-115–120	Canonical pair identity, identical operator, and evidence semantics	Exact raw-pair identity; paired impulse/state/sampling equality; zero cross branches; retained $E_x/E_r/E_{xr}$ origin, support, uncertainty, protection, and direct/inferred causes; cause-preserving hard-union fixtures.
REQ-121–125	Protection, symmetric pair action, and immutable lifecycle	Expected-source/leakage protection cases; direct- r -to-inferred- x and converse hard events; complete Resolve trace; Apply mutation rejection; successor-attempt identity.
REQ-126–130	Paired level shifts and spectral admission	One event/support/reset with unequal coordinate amplitudes and asymmetric correction availability; no copied amplitude; r -only candidate rejection; complete shared-notch admission and identical response.
REQ-131–135	Donor exception, grid, provenance, and uncertainty	No- r -donor rejection; full causal unavailable influence through state/filter support; exact common grid; reconstructible cause/action record; conditional versus selection/parameter/correction uncertainty and cross-block retention.
REQ-136–138	Product naming and package boundaries	Canonical-pair versus unrequested/globally-unavailable- r and descendant identity; x -only CAL; no automatic $r \rightarrow x$ correction; unchanged producer facts across PTC/VAL consumer policies.
REQ-139–143	R0.12 pair identity and support corrections	Native-IQ mapping then single ALIGN; pair-based learning/evaluation/replay from $u^{(0)}$; separate common grid, pair-action support, coordinate availability, and covariance support; failed affine-correction influence through FIR/IIR/sampling; immutable raw- r event evidence after conditioned- r correction.

C Informative response and representation checks

Review the response at the same granularity as the realized operation. A stored coefficient vector can support a component-level check; it does not by itself establish complete response. For each coherent segment, trace the ordered factorization, exact coefficients and precision, mask/state/edge exceptions, detector mixing, sampling phase, and support expansion to the artifact presented to consumers. Mark complete response unavailable if any enabled response-changing stage cannot be represented truthfully.

Begin that trace with exact general IQ-to- x/r mapping authority, one upstream ALIGN relation, and the admitted aligned RTC boundary. Reject a second ALIGN application in the RTC-local operator; compose it only for an explicitly identified end-to-end response. Exercise the common paired operator while retaining raw- r parentage and requested same-grid conditioned r . Verify identical ordinary paired stages, zero fixed-state cross branches, retained r -origin effects on pair-plan selection/support, no x -donor branch in the r response, honest local/global r unavailability, and full cross-coordinate selector and uncertainty dependence. Record the common index grid, common pair-action/operator support, each coordinate’s numerical availability, and joint covariance support as separate objects. Verify ordinary reset at level shifts and admit carry only under separate continuity authority with response and residual criteria. Confirm that atmospheric templates change only diagnostic or declared selector state, not numerical science x . Reject conditioned r that has a separate lifecycle, mismatched grid, invented repair, missing response/availability facts, numerical cross mixing, or corruption of valid x merely because conditioned r is unavailable.

Use the prediction IDs rather than inventing new expected science after observing output. Interior impulses and sinusoids exercise fixed linear components; donor impulses exercise detector mixing; split-state fixtures exercise recurrence; noncenter causes exercise transitive influence; and odd/even factor enumerations exercise phase-zero identity. These checks are complementary. An interior transfer match cannot close edges or selection, and an output-value match cannot close identity, response, or provenance.

Representation fidelity is a separate record. Reconstruct the exact operator and exceptions from persisted factors, intervals, mappings, and state. Verify that omitted detail is typed optional and inert. A readable plot or plausible header is not proof that the stored representation is complete.

D Informative state and failure checks

Exercise each semantic state independently: not requested, disabled, automatic, rejected, invalid, failed, unavailable, unsupported, and omitted. Also separate original validity, synthesis, replacement, mask membership, numerical finiteness, edge/context, response availability, eligibility, and uncertainty. A fixture that combines every failure into

NaN or one generic bit cannot demonstrate these distinctions.

Additionally separate pair-malformed, x -invalid, r -invalid, mapping-unavailable, leakage-unavailable, level-shift-candidate, level-shift-accepted, physical-transition-unmodeled, propagated-transition-influence, additive-offset-applied, additive-offset-unavailable, segment-interior, application-specific plateau-admissibility, plateau-advisory, reset, atmospheric-diagnostic-unavailable, and conditioned- r not-requested, locally unavailable, globally unavailable, and available states. Confirm that missing one raw pair member fails required admission while coordinate-specific evidence and numerical validity remain expressible, pair-level pathology support is conserved, invalid r slots remain on-grid, and no r failure changes valid conditioned x .

For each required output member, inject a bounded write or authority failure and verify atomic completion behavior. Inspect that no prior-observation factor, donor, state, mask, rate, product link, or completion marker remains. Then toggle the complete spike-event/donor manifest and other optional inert diagnostics; the scientific and numerical bundle, compact treatment state, and spike-population summary must remain identical apart from the permitted optional detail. Separately verify that selected despiking actually modifies accepted target x cells according to the resolved recovery policy.

Every learned operation requires an additional stop check before apply. The review must bind the declared learning population and every numerical predicate to approved authority, show the complete candidate disposition, and preserve learned-evidence and resolved-plan state. If the resolved plan is absent or incompatible, apply does not begin. A native-cadence result produced after silent fallback is not evidence of an applied learned plan.

Bounded iterative notch refinement requires one such record per attempt. Trace the attempt and current accepted-plan indices, original admitted pair $u^{(0)}$, learning parent/population, prior accepted plan, complete proposal, typed disposition, and any new immutable evaluation product, together with candidate dispositions, response and science budgets, and continuation or stop cause. Confirm that the final plan is replayed against $u^{(0)}$ unless a separately authorized cascade is recorded. Rejected attempts do not create accepted plans or evaluation products. The record distinguishes actual attempts, configured maximum, and accepted plans; an early stop has no artificial no-op attempts. Maximum-attempt, oscillating, and nonconvergent states are not convergence.

E Informative completion checklist

- Every SCI-RTC-REQ identifier has one explicit disposition and evidence record; ranges are not used as a substitute for per-ID closure.
- Every applicable SCI-RTC-PRED identifier has a preregistered expected result and tolerance, plus observed values and messages.
- Every open owner decision is either resolved by approved authority or blocks the named operation/claim exactly as the ledger states.
- Every RTC application context retains exact paired raw x/r identity and independent validity; conditioned x is the required numerical RTC output, raw r parentage and causal uses remain in the bundle, and SCI-CAL consumes only x while binding the complete consumer-neutral RTC parent.
- Application context, immutable resolved plan, and realized record are distinct one-way objects; context labels neither partition admitted operation classes nor change the atomic bundle schema.
- Missing either pair member fails admission; no reorder, independent join, copy, synthesis, zero fill, r calibration, r -to- x correction, or r -as- x donor route is accepted.
- Direct- x , direct- r , and joint evidence retain origin and causal class; accepted hard support is unioned once at Resolve, inferred pair consequences are not reported as direct detections, and Apply does not mutate the plan.
- Every ordinary canonical stage has exactly identical x/r masks, segments, guards, coefficients, state/reset, phase, sampling, grid, and self-response, with both fixed-state cross-coordinate branches zero.
- The common index grid, common pair-action/operator support, coordinate-local numerical-availability sets, and joint covariance support are separately represented; local unavailability never deletes or reindexes a paired occurrence.
- Expected r optical or atmospheric response is protected evidence, not automatic pathology; candidate spectral evidence never becomes shared subtraction without the complete selected admission budgets.
- The owner-modified raw donor ratio is checked in both direction and availability cases, with no dependency on legacy responsivity.
- Direct synthesis/replacement is universally excluded, while noncenter full-support influence is preserved and evaluated by each declared downstream consumer policy.

- Complete response includes every enabled stage or is typed unavailable.
- Leakage diagnostics preserve separate atmosphere and bright-source parents, estimator bias controls, scalar/frequency status, support, uncertainty, and pre/post response; agreement is not automatic promotion.
- Level-shift learning uses original paired data with isolated-spike candidates masked or downweighted; donor replacement begins only after segmentation, remains within stable segments, and never crosses a shift.
- Selected despiking modifies accepted target x cells. Normal output retains compact treatment state and useful spike-population counts and characteristics; a complete event/donor manifest is optional diagnostic detail and its toggle is numerically inert.
- Every accepted shift has paired amplitudes or typed unavailability, transition mask, guards, segment identity, reset/carry state, application-specific plateau support, selected additive correction only with sufficient stable support, persistent transition flags, and no gain- or responsivity-change model.
- Paired shifts have one event/support/segmentation/reset and independent coordinate-specific affine amplitudes, uncertainty, and correction availability; neither amplitude is copied or mixed. An unavailable required affine correction propagates through the complete downstream FIR, IIR-to-reset, and sampling influence for that coordinate only, while raw- r parentage and original event evidence remain immutable.
- The x donor exception creates no r value and makes requested conditioned r and its response unavailable over the full transitive influence while preserving the exact grid and all causes.
- Atmospheric templates are diagnostic evidence only and do not alter science x ; numerical common-mode removal requires separate authority.
- A conditioned r output is unavailable unless its own operator, response, support, unit, validity, uncertainty, and provenance are bound.
- Requested, effective, observation-resolved, learned-evidence, resolved-plan, realized, and immutable product/stage substates remain distinct within the context-plan-record lifecycle.
- Every covariance or uncertainty claim explicitly states its included and excluded components and correlations; excluded or unknown terms are not silently zero, and only declared complete coverage is called total.
- Each learned operation names its learning population, and apply rejects any mutation of the resolved plan; online adaptation is a separate unavailable or separately authorized estimator.
- Filter validation varies scientific design choices over the admitted scan, projected-beam, and source domain; coefficient fixtures alone do not establish science qualification.
- Iterative notch refinement has a finite configured maximum, distinct actual-attempt and accepted-plan counts, no artificial no-op attempts, complete cumulative proposals, immutable accepted-plan apply, explicit original-pair $u^{(0)}$ replay or cascade authority, and a typed stop or nonconvergence disposition.
- Required failures are atomic; optional detail is inert; unexpected error-level messages are zero in any run claimed successful.
- The final report names exactly one supported claim layer and explicitly withholds all stronger layers.

This checklist does not itself establish any item. It is a review index for future evidence governed by the shared core.